

Williams®

16P-3001-101
February, 1981

 Williams
DEFENDER



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Williams® 
ELECTRONICS, INC.
3401 N. California Ave. Chicago, Il. 60618
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(312) 267-2240



CPU/Video Board

Note modification required on all sample games described in Service Bulletin V-1. Should failure of 4D 7486 IC occur, make foil cuts on board as described in Service Bulletin V-3.

ROM Board

When blue-labeled ROMs are used, ROM 4 is not used, jumpers W2 and W3 are connected, and jumper W1 and W4 are removed. When green-labeled ROMs are used, ROM 4 is used, jumpers W1 and W4 are connected, and jumpers W2 and W3 are removed. Between blue and green-labeled ROM sets, ROMs 2, 6, 7, 8, 9, 10, 11, and 12 are compatible.

Note foil cuts and addition of resistors required as described in Service Bulletin V-2.

Interface Board

Note foil cuts and addition of resistors required as described in Service Bulletin V-2.

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SERVICE BULLETIN

DEFENDER CPU/VIDEO BOARD MODEL D8356

Watchdog Circuit Modification

INTRODUCTION

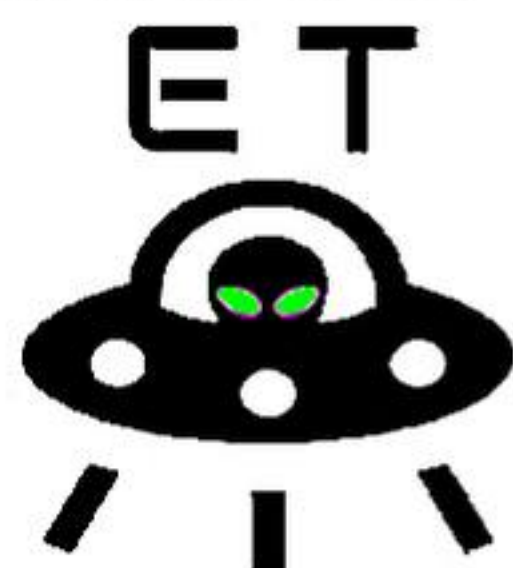
The Watchdog timer on D8356 CPU/Video Board in sample games has exhibited a tendency to timeout too quickly. This results in the game intermittently resetting. Should this occur with credits in the game, the credits will be removed.

MODIFICATION

All CPU/Video Boards in sample games shipped prior to 12-8-80 must be modified as follows. The modification consists of cutting a trace and adding a jumper. Refer to the drawing on the other side of the sheet and proceed as follows:

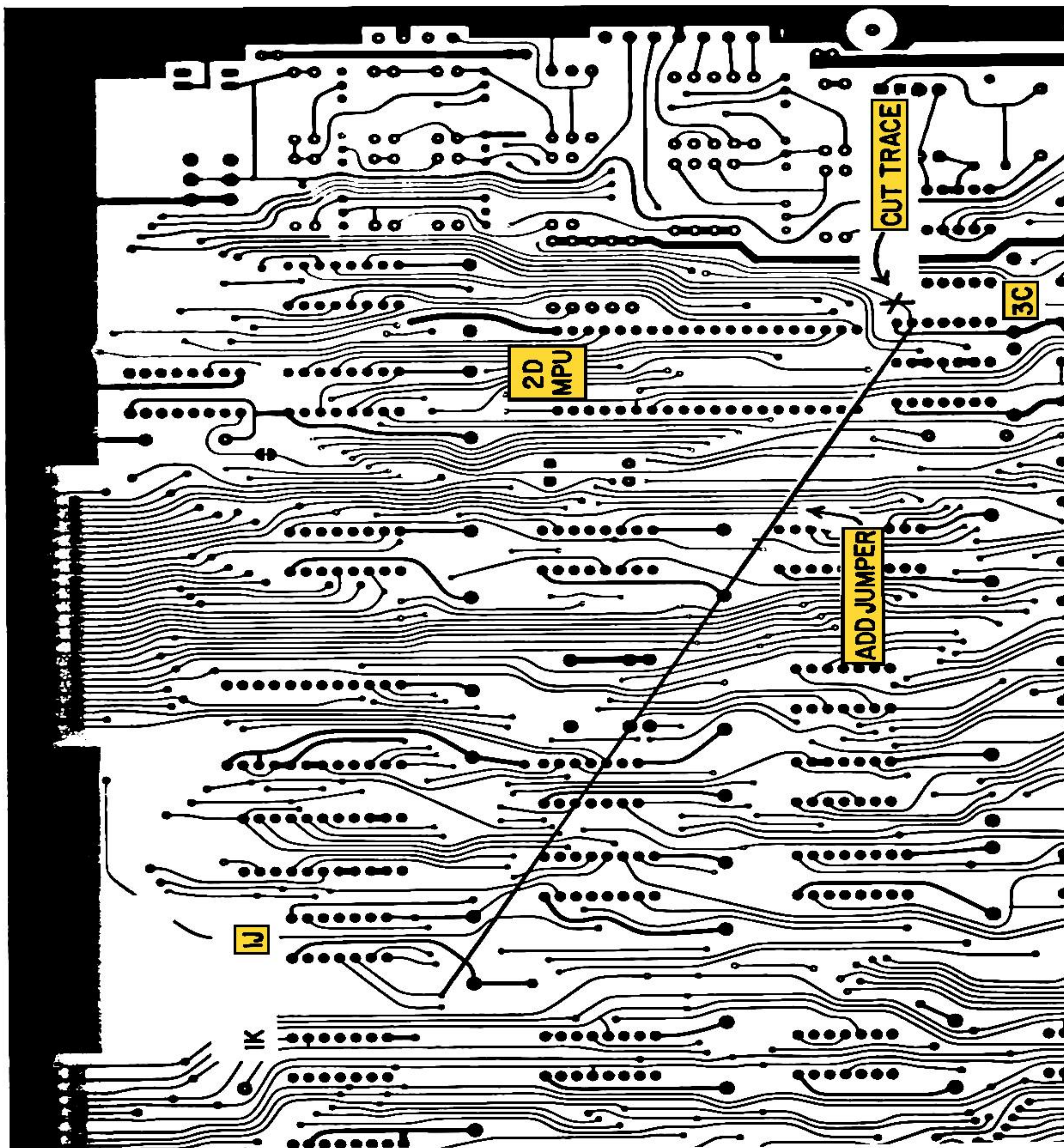
1. Locate the trace from 3C pin 13 on the solder side of the board. 3C is a type 74393 or 74LS393.
2. Cut the trace located in step 1.
3. Obtain a 6 inch length of 30-gauge or other available gauge of insulated wire.
4. Strip back the insulation at each end of the wire approximately 1/8-inch and tin the wire.
5. Solder one end of the wire to 3C pin 13 on the solder side of the board.
6. Locate the trace connected to 1J pin 10. 15 is a type 7404.
7. Solder the other end of the wire to the feedthru connected to 1J pin 10.

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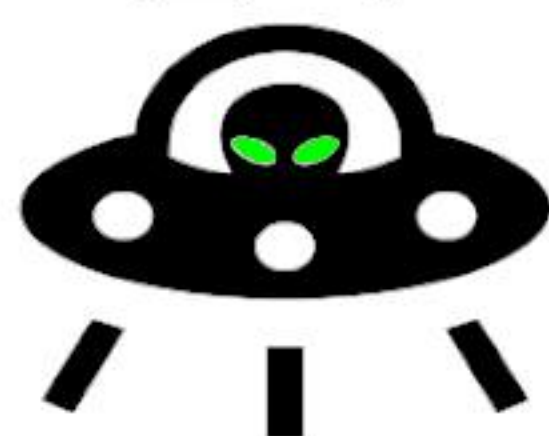
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SERVICE BULLETIN

DEFENDER ROM AND INTERFACE BOARDS MODELS D8357 AND C8358 Addition of Resistors

INTRODUCTION

Switch inputs to the ROM Board and Interface Board are applied to CMOS hex inverters which are susceptible to possible damage from static discharge. Addition of series resistors is required to provide protection.

MODIFICATION

This modification must be made to these boards in all sample games. Refer to the drawings on the reverse side and proceed as follows:

1. Obtain 23, 1K $\pm 10\%$, 1/4 W resistor (Williams Part No. 5A-8984)
2. Cut the traces as indicated in seven places on the ROM board.
3. Solder in a resistor bridging each cut trace in the pads on the board.
4. Repeat steps 2 and 3 for the Interface board, cutting 16 traces.

ENHANCED BY:



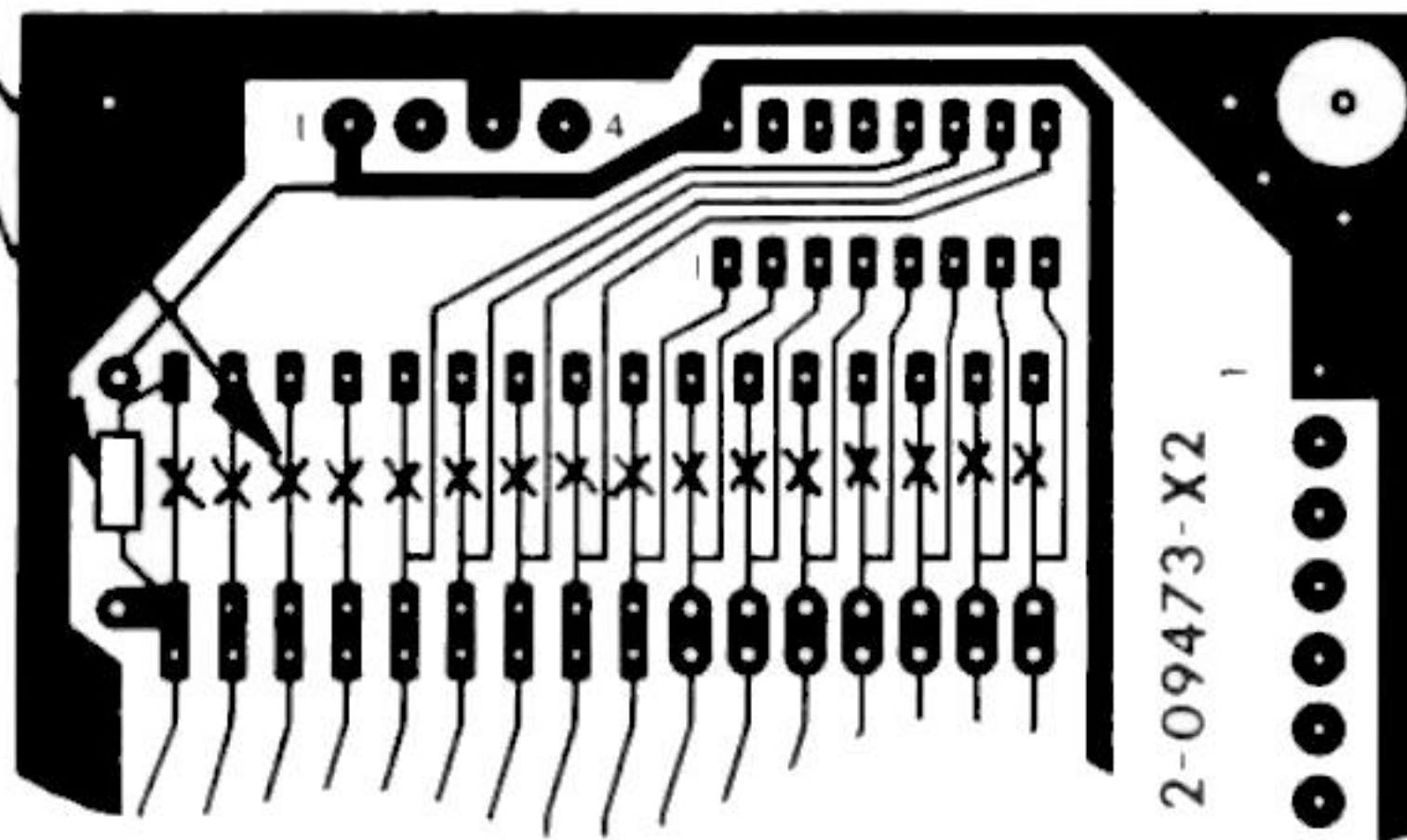
5



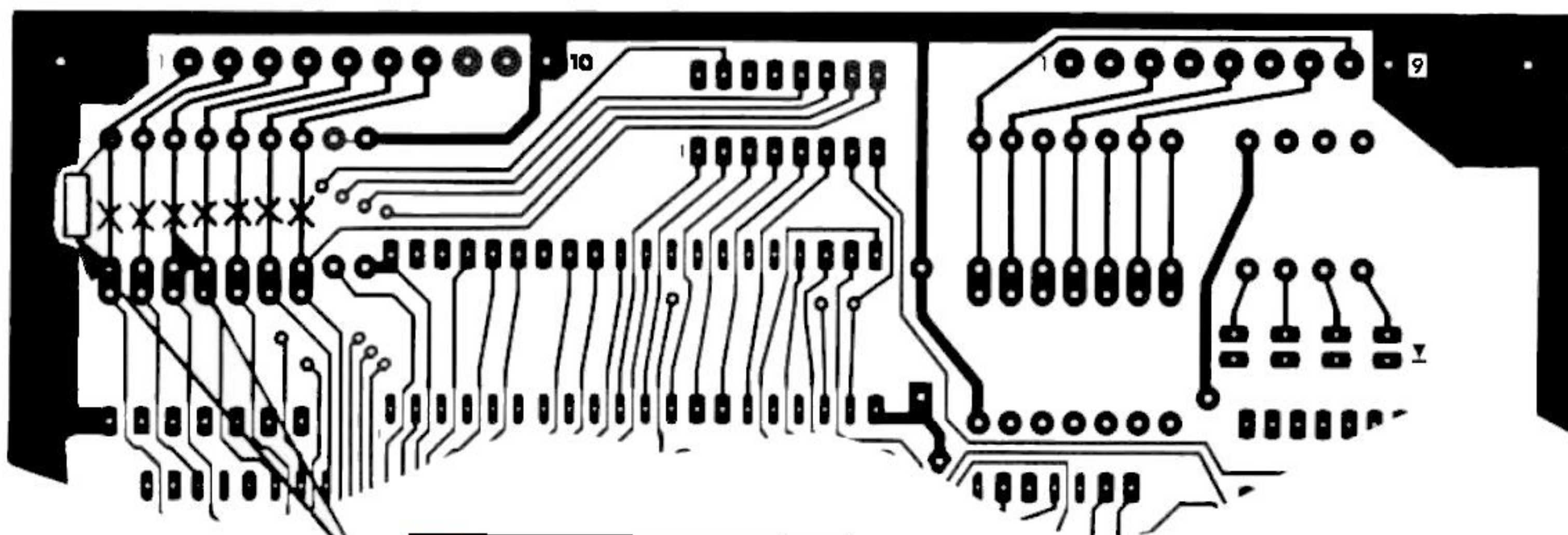
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CUT 16 TRACES,
ADD 16 RESISTORS

C 8358
INTERFACE BOARD



D 8357
ROM BOARD



CUT 7 TRACES,
ADD 7 RESISTORS

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SERVICE BULLETIN

DEFENDER CPU/VIDEO BOARD
MODEL D8356
Foil Cuts at 4D 7486 IC

INTRODUCTION

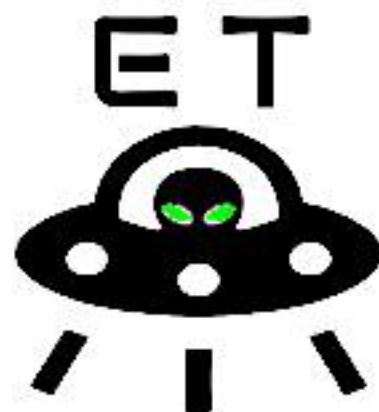
Termination of pins for unused gates in the 4D 7486 IC may cause overheating in ICs with certain pin configurations. This could cause the IC to fail.

MODIFICATION

Failure will normally occur within a few days of operation so it is recommended not to make the modification unless replacement of the 4D 7486 IC is required. When replacing the chip, proceed as follows:

1. On the foil side of the board, remove solder from pins 6 and 7 and pins 8 and 9 of the 4D IC.
2. Cut the foil between pins 6 and 7 and between pins 8 and 9.
3. Carefully resolder pins 6, 7, 8, and 9, making sure that solder does not bridge across the pins.

ENHANCED BY:

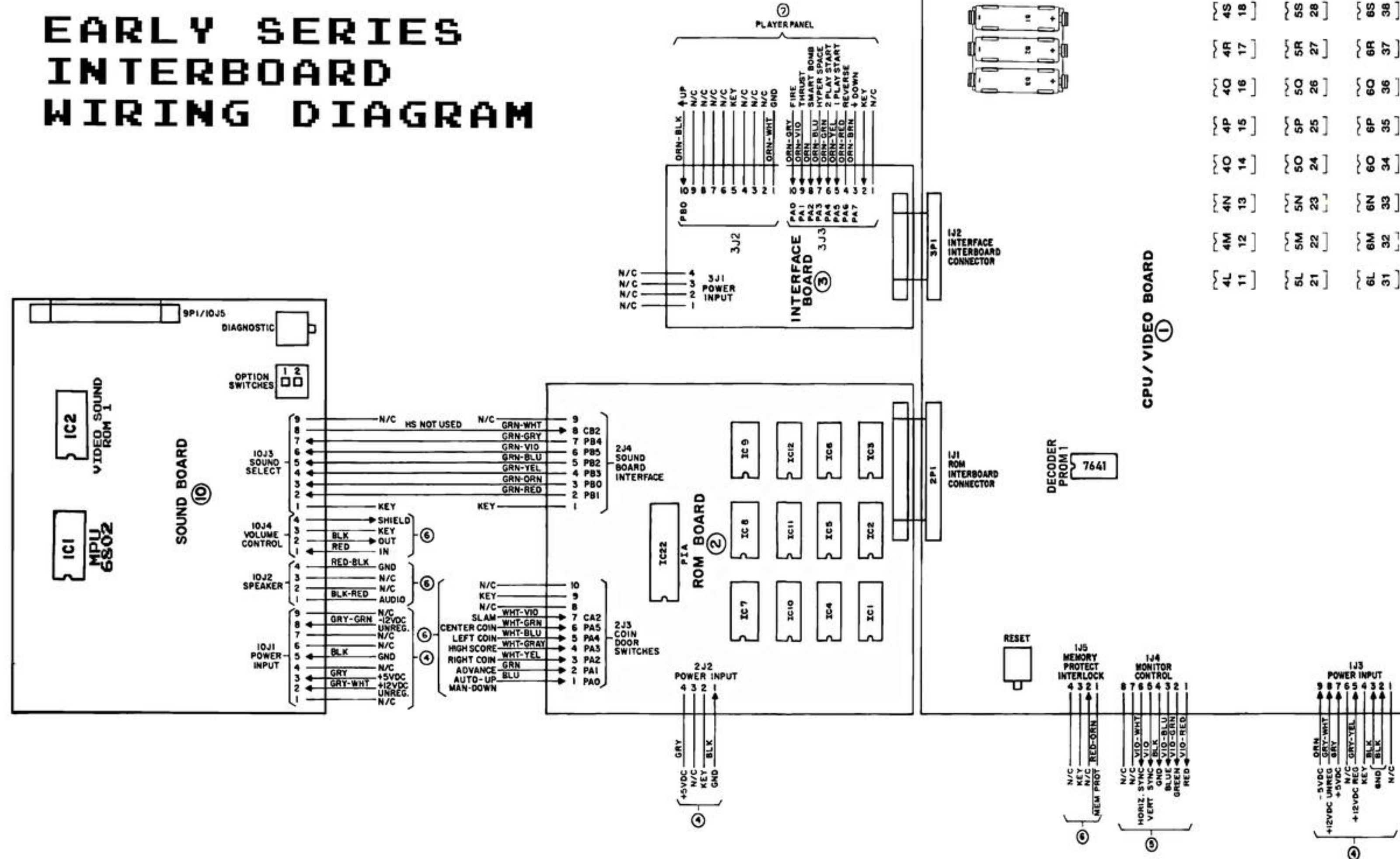


7-8

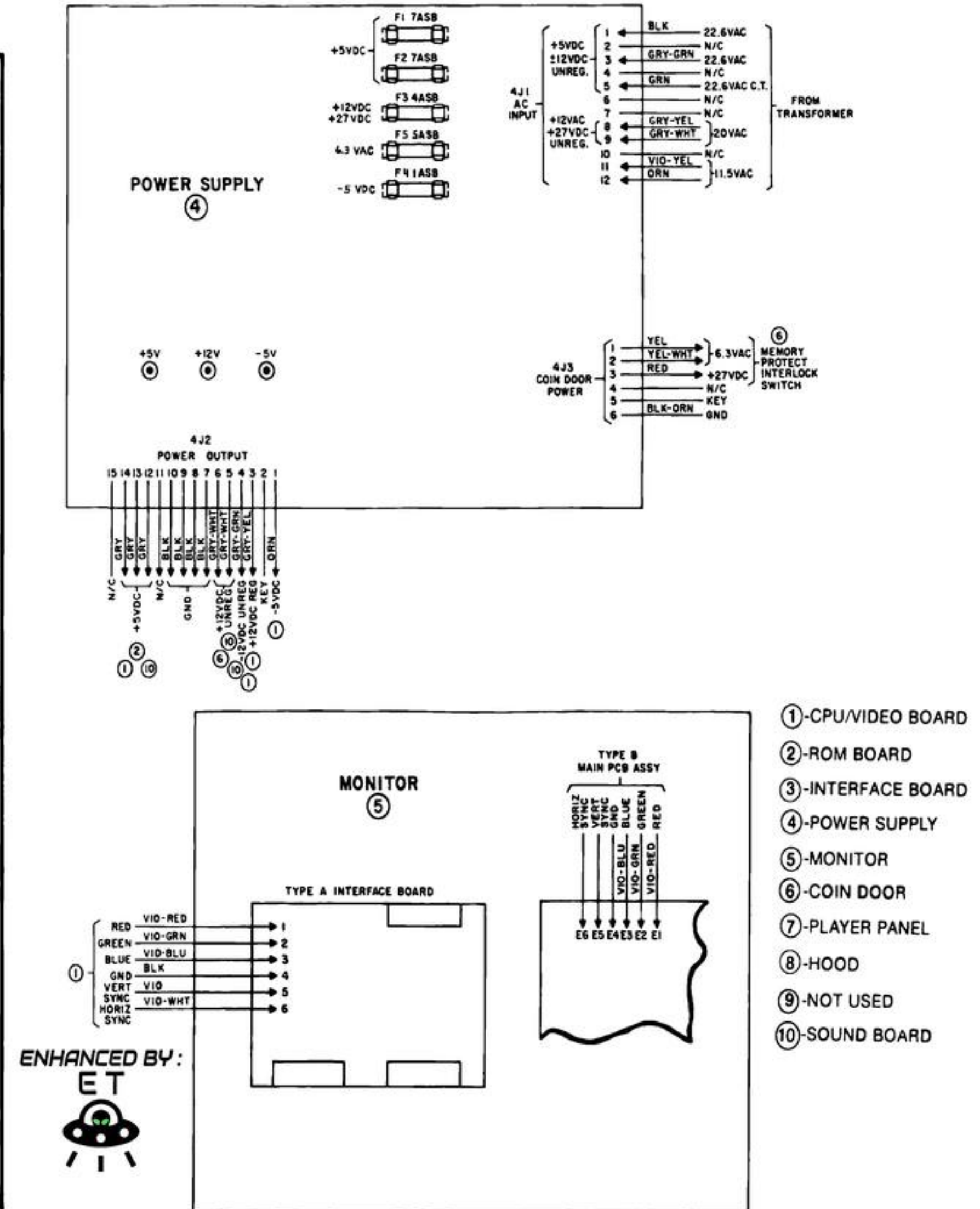


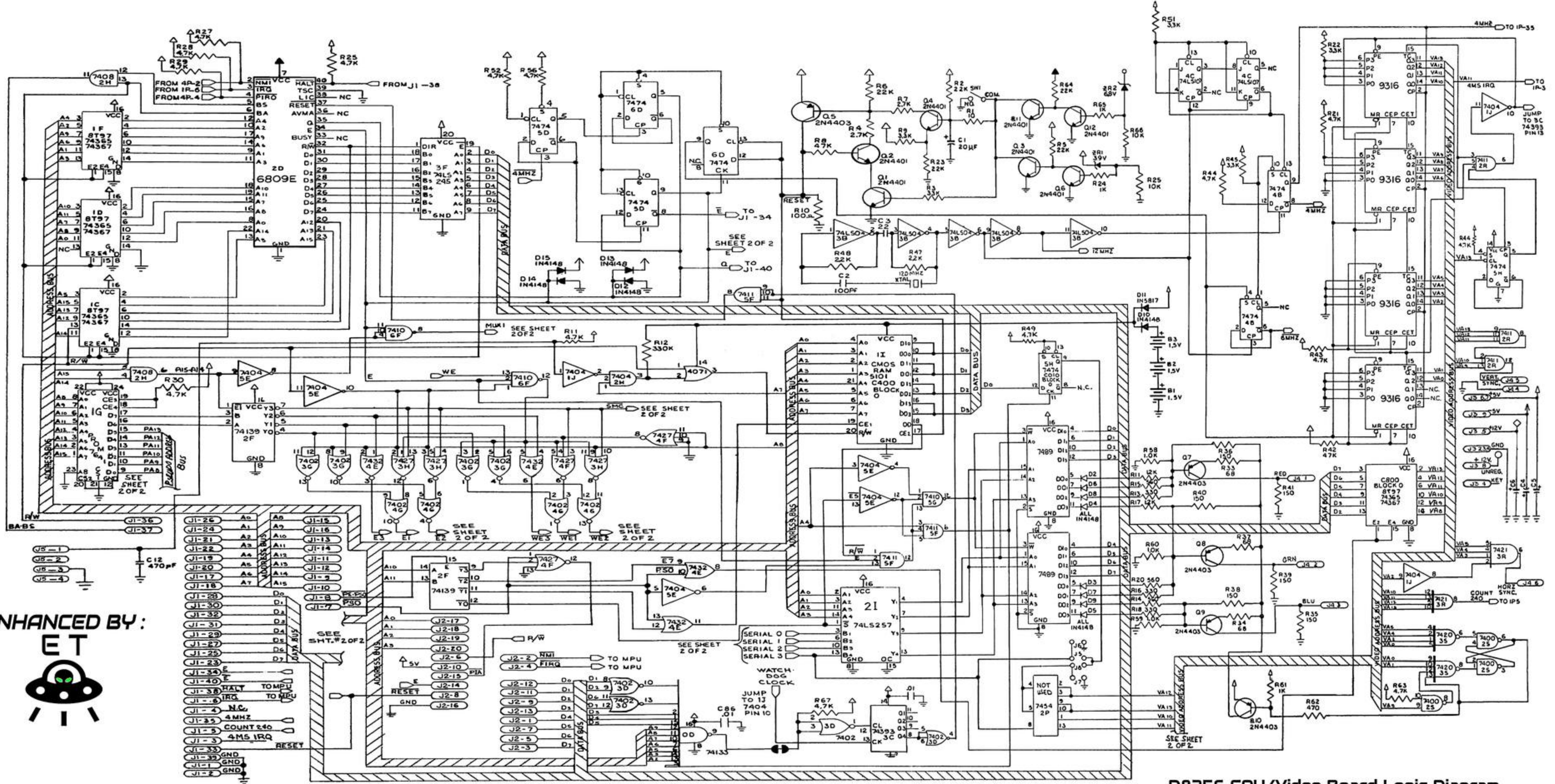
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EARLY SERIES INTERBOARD WIRING DIAGRAM



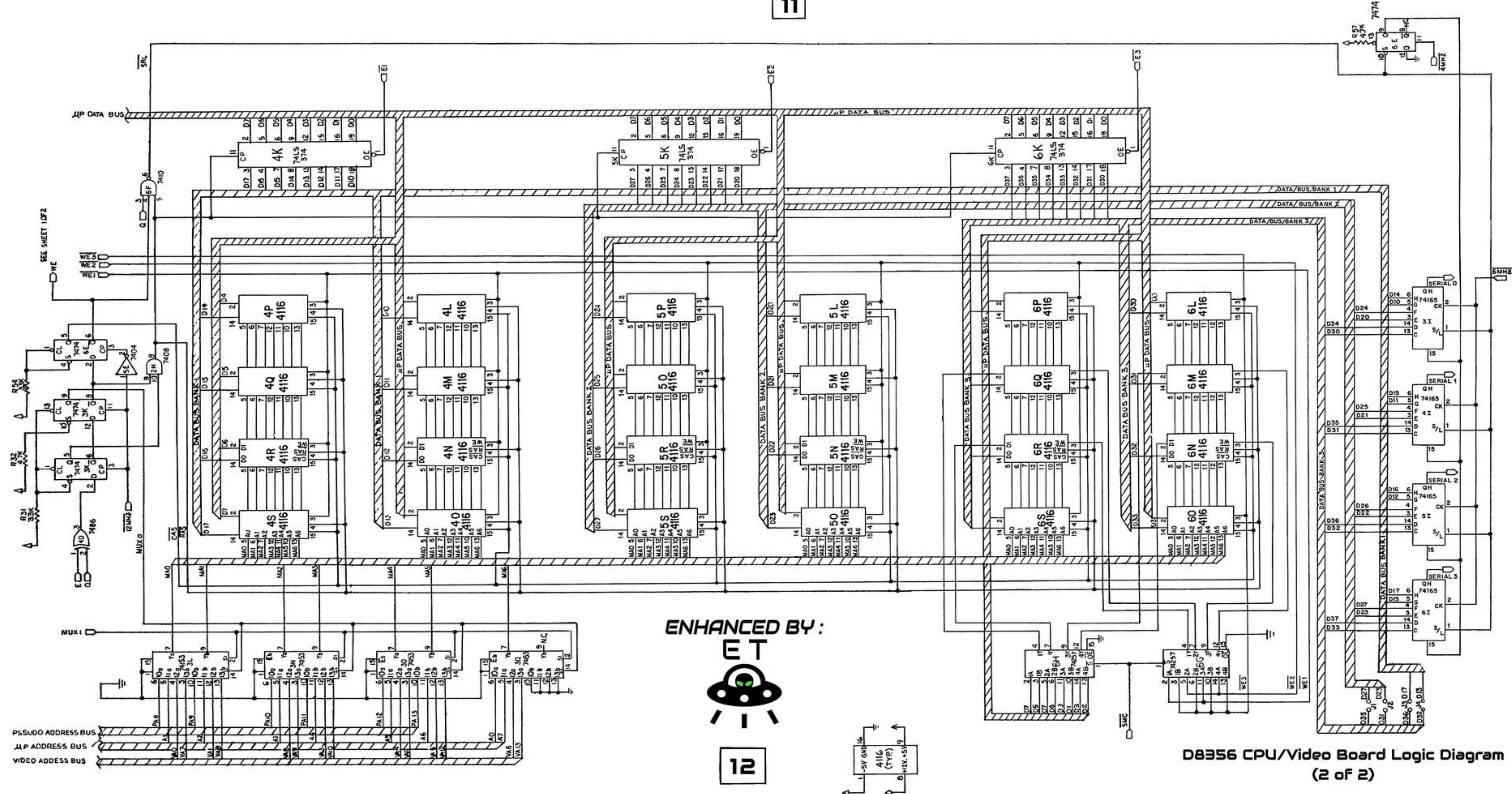
SIDE PANEL





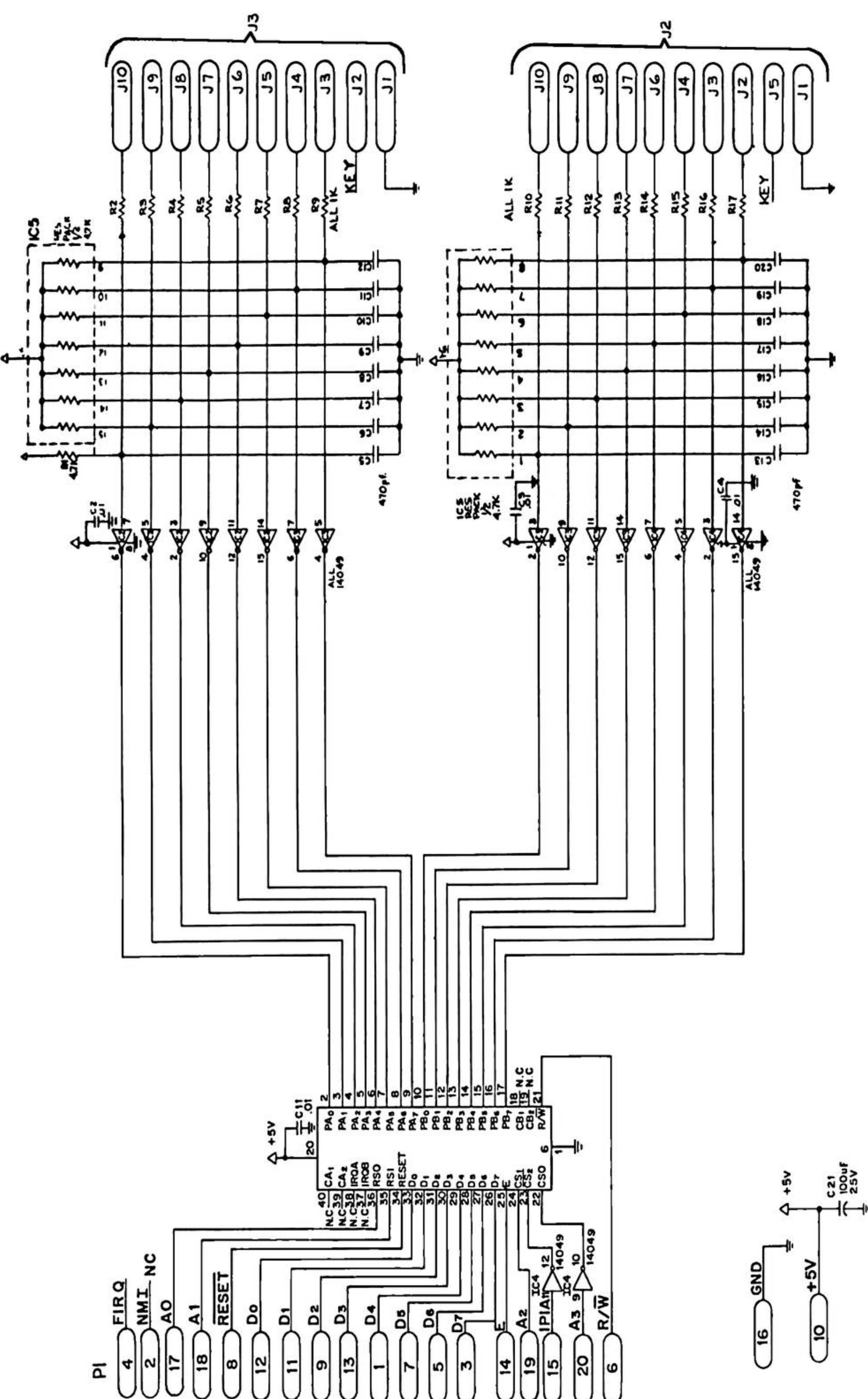
D8356 CPU/Video Board Logic Diagram
(1 of 2)

11

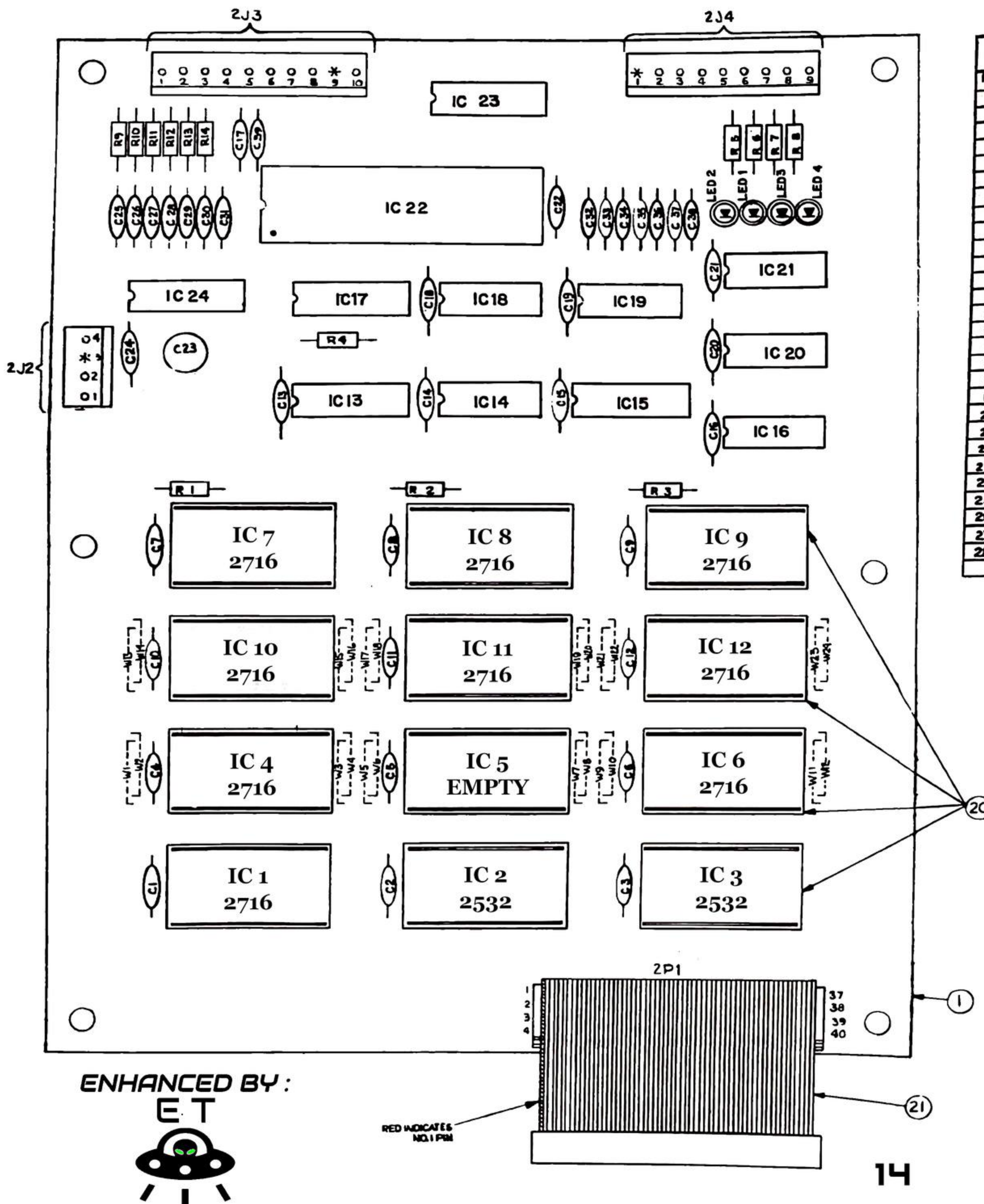


D8356 CPU/Video Board Logic Diagram
(2 of 2)

12

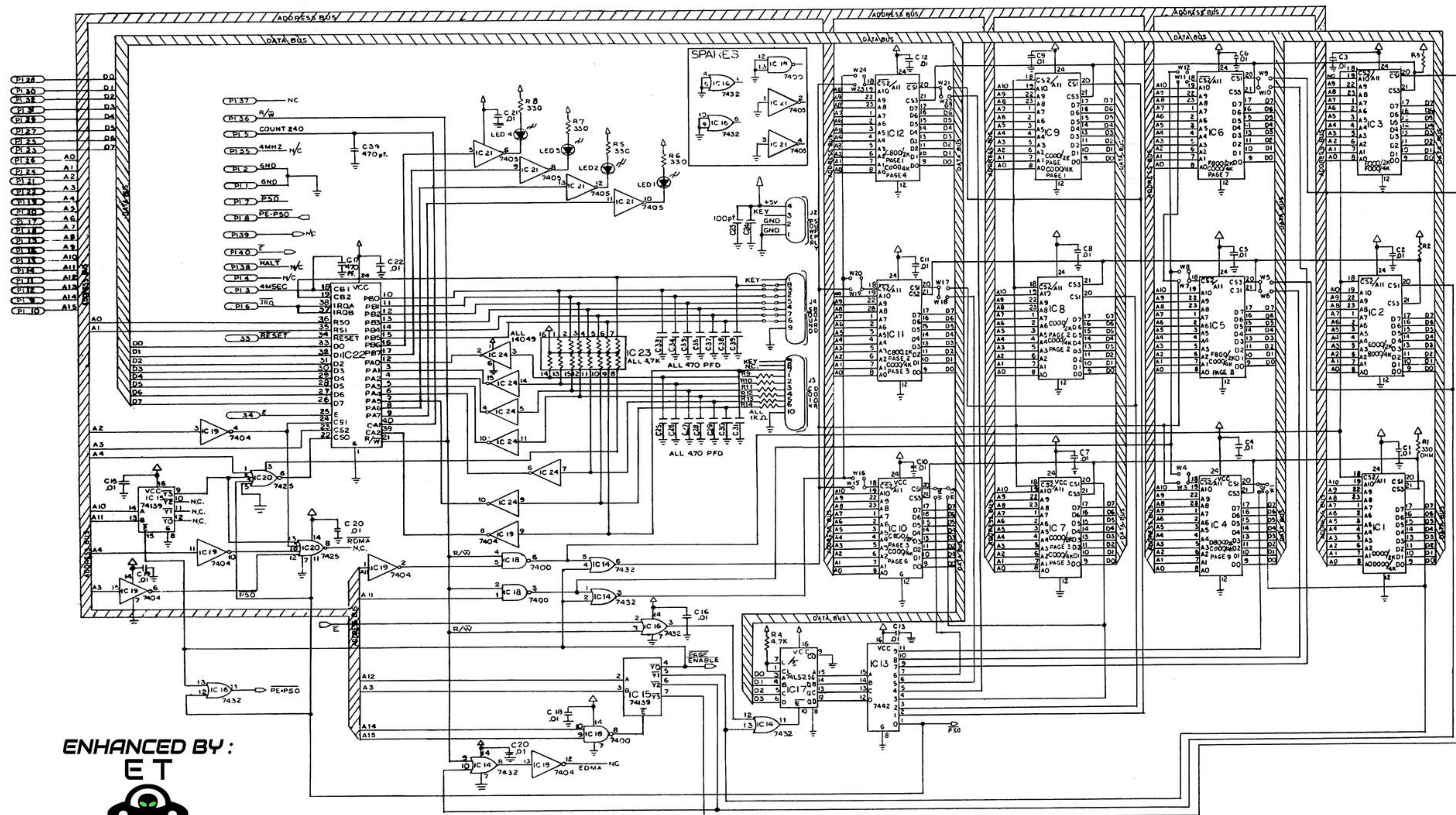


BILL OF MATERIAL				
ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQD NO.
1	5712-09473-00		BARE P.C. BOARD	1
2	5310-08373-00	IC 2, IC 3, IC 4	4049, INV. HEX BUFFER	3
3	5018-Q5239-00	IC 5	4.7 K 15 RES, 16 DIP	1
4	5010-08991-00	R1	4.7K Ω RESISTOR, $\frac{1}{4}$ W 5%	1
5	5043-08980-00	C1, C2, C3, C4	.01 MFD. CAPACITOR 50V. AXIAL, CERAMIC +80-20%.	4
6	5043-09063-00	C5 THRU C20	470 PFD. CAPACITOR 50V. AXIAL, CERAMIC	16
7	5040-09421-00	C21	100 MFD. CAPACITOR 25V. RADIAL, ELECTROLYTIC	1
8				
9	5791-09444-00	J2 J3	10 PIN HEADER, MOLEX 09-65-1101	2
10	5793-09540-00	3P1	20 PIN RIBBON CABLE ASSY.	1
11	5430-08972-00	IC1	MC6821 PIA	1
12	5010-09358-00	R2 THRU R17	1 K OHM RESISTOR $\frac{1}{4}$ W 5%	16



BILL OF MATERIAL				
ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	NO. REQ'D
1	5771-09472-00		BARE PC BOARD	1
2	5180-09013-00	IC 13	7442 BCD DECODER	1
3	5180-09477-00	IC 14, IC 16	7432 QUADRUPLE 2 INPUT	2
4	5181-09344-00	IC 15	74LS139 DUAL 2 TO 4 LINE DECODER	1
5	5181-09501-00	IC 17	74LS395 4-BIT UNIVERSAL SHIFT	1
6	5180-09073-00	IC 18	7400 QUAD 2 INPUT NAND GATE	1
7	5180-09013-00	IC 19	7404 HEX, INVERTER	1
8	5180-09381-00	IC 20	7425 NOR GATE WITH STROBE	1
9	5180-09349-00	IC 21	7405 OPEN COLLECTOR OUTPUT	1
10	5430-09772-00	IC 22	MC 6821 PIA	1
11	5019-09239-00	IC 23	18 PIN, 15 RES DIP PACK 4.7K OHM	1
12	5671-09019-00	LED 1 THRU LED 4	LED, RED	4
13	5040-09442-00	R1 THRU R3, R5-R8	RESISTOR, C.F. 330 OHM 5% 1/4 W	7
14	5040-09591-00	R 4	RESISTOR, C.F. 4.7K OHM 5% 1/4 W	1
15				
16	5043-09380-00	C1 THRU C16, C18	CAR CERAMIC, 10MFD, 50V, AXIAL	21
17	5040-09421-00	C 23	CAP. ELECT., 100 MFD, 25V, RAD.	1
18	5043-09396-00	C 24	CAR CERAMIC, 1MFD, 50V, AXIAL 220%	1
19	5043-09049-00	C17, C18 THRU C24	CAR CERAMIC, 470 PFD, 50V, AXIAL	18
20	5700-09004-00		24 PIN IC SOCKET	12
21	5799-09353-00	2P1	40 PIN RIBBON CABLE ASSEM	1
22	5791-09028-00	2J2	4 PIN MALE CONNECTOR 09-65-1041	1
23	5791-09444-00	2J3	10 PIN MALE CONNECTOR 09-65-1011	1
24	5791-09027-00	2J4	9 PIN MALE CONNECTOR 09-65-1051	1
25	5310-08975-00	IC 24	4049 INVERTING HEX. BUFFER	1
26	5010-09358-00	R9 THRU R14	RESISTOR, C.F. 1K OHM 5% 1/4 W	6
27	SEE CHART	IC1 THRU IC12	SEE CHART	AS REQ'D
28	5010-09534-00	W1 THRU W24	RESISTOR, 0 OHM	AS REQ'D

GAME NAME	'DEFENDER' RED LABEL	DEFENDER BLUE LABEL	DEFENDER GREEN LABEL	DEFENDER WHITE LABEL	
ASSEM. NO.	D-8572-3001	D-8520-00	D-8520-00		
IC 1	A-5343-09636	5343-09595-00	5343-09821-00		
IC 2	A-5343-09637	5343-09596-00	5343-09596-00		
IC 3	A-5343-09638	5343-09597-00	5343-09622-00		
IC 4	A-5343-09639	NOT USED	5343-09623-00		
IC 5	NOT USED	NOT USED	NOT USED		
IC 6	A-5343-09640	5343-09600-00	5343-09600-00		
IC 7	A-5343-09641	5343-09601-00	5343-09601-00		
IC 8	A-5343-09642	5343-09602-00	5343-09602-00		
IC 9	A-5343-09643	5343-09603-00	5343-09603-00		
IC 10	A-5343-09644	5343-09604-00	5343-09604-00		
IC 11	A-5343-09645	5343-09605-00	5343-09605-00		
IC 12	A-5343-09646	5343-09606-00	5343-09606-00		
JUMPERS	W2, W4, W6, W7, W9, W11, W13, W16, W17, W20, W21, W24	W2, W3, W6, W7, W10, W11, W13, W16, W17, W20, W21, W24	W1, W4, W6, W7, W10, W11, W13, W16, W17, W20, W21, W24		



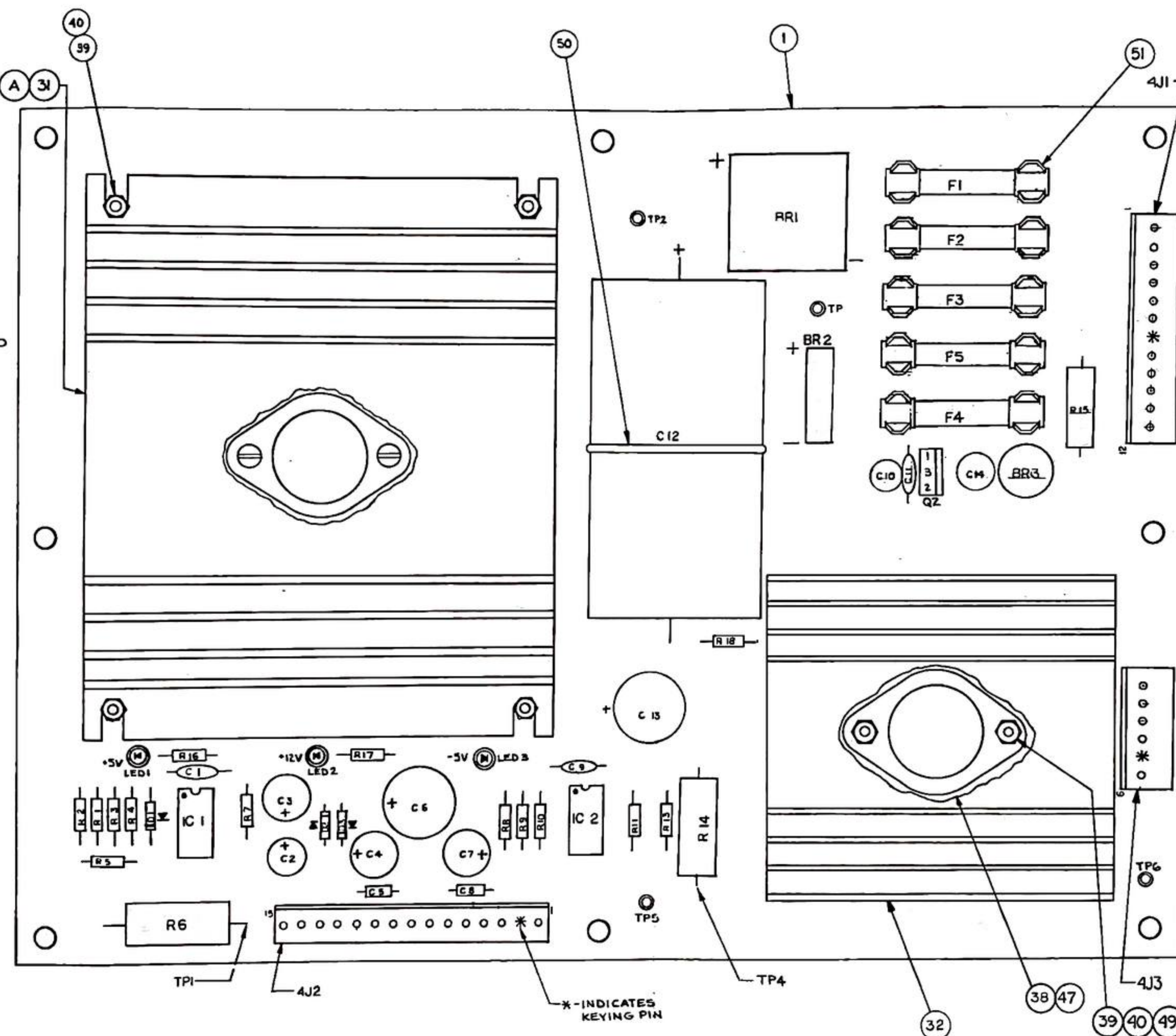
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BILL OF MATERIAL			
ITEM NO	PART NO	DESCRIPTION	REQ'D NO
1A	5162-09423-00	TRANSISTOR NPN 2N6057	1
2A	5700-09435-00	TRANSISTOR-SOCKET TOS	1
3A	5709-09431-00	HEATSINK	1
4A	20-3729-00	THERMAL COMPOUND 1.5 OUNCE	1
5A	5701-09333-00	TOS 3 MIL MICA INSULATOR	1
6A	4006-01003-B	MS-6-22x1/2-P-PH	2
7A	5853-09591-00	LEAD WIRE #18 AWG 3/8 STRIPPED (GRN)	1
8A	5853-09592-00	LEAD WIRE #18 AWG 3/8 STRIPPED (WHT)	1
9A	5853-09593-00	LEAD WIRE #18 AWG 3/8 STRIPPED (YEL)	1

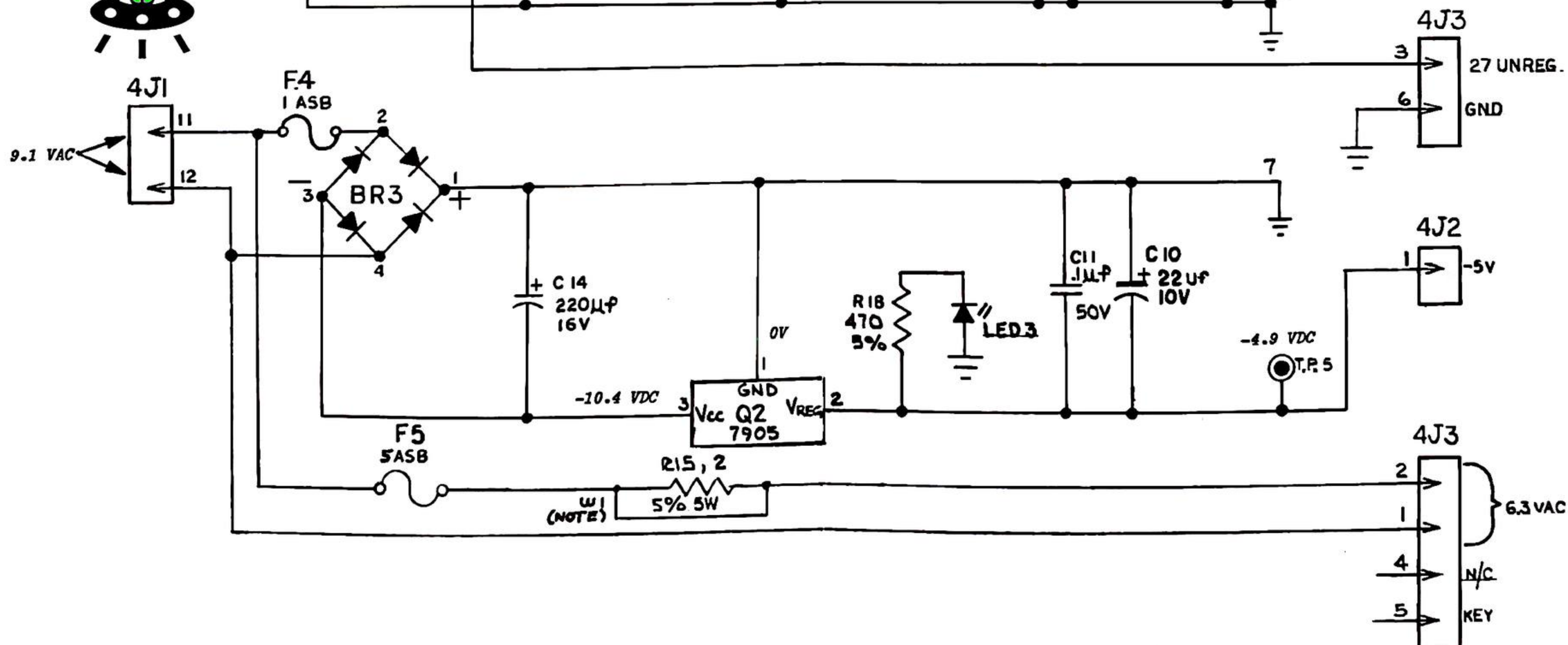
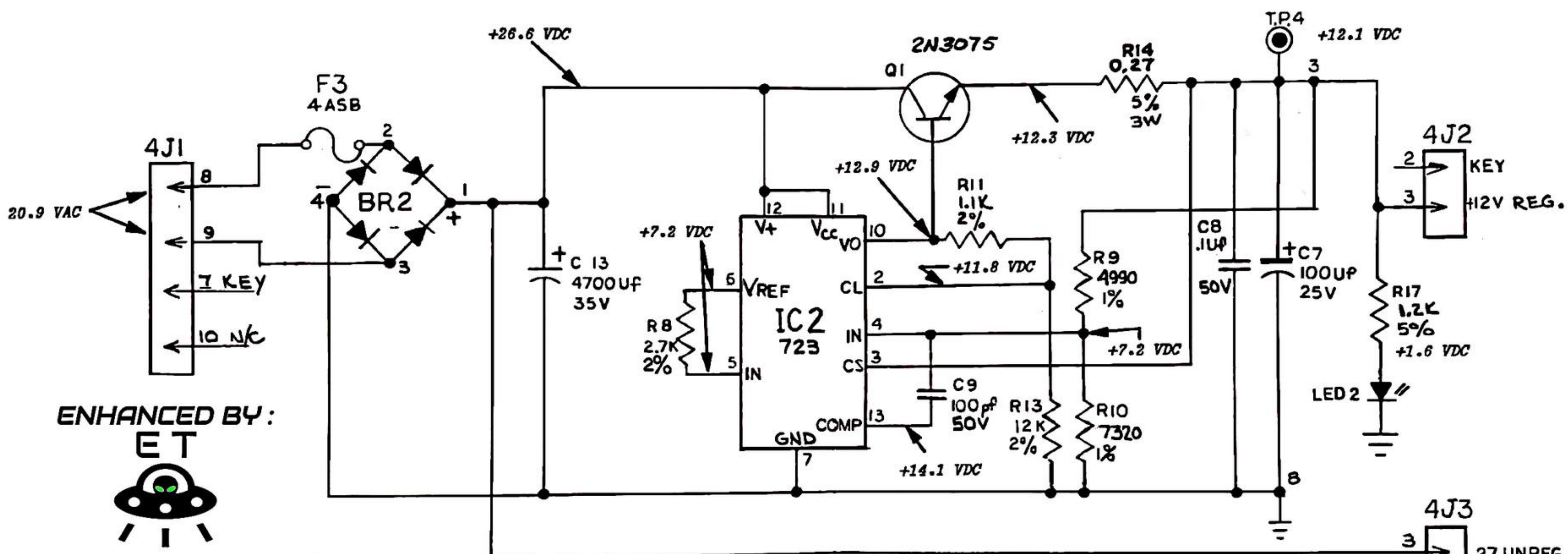
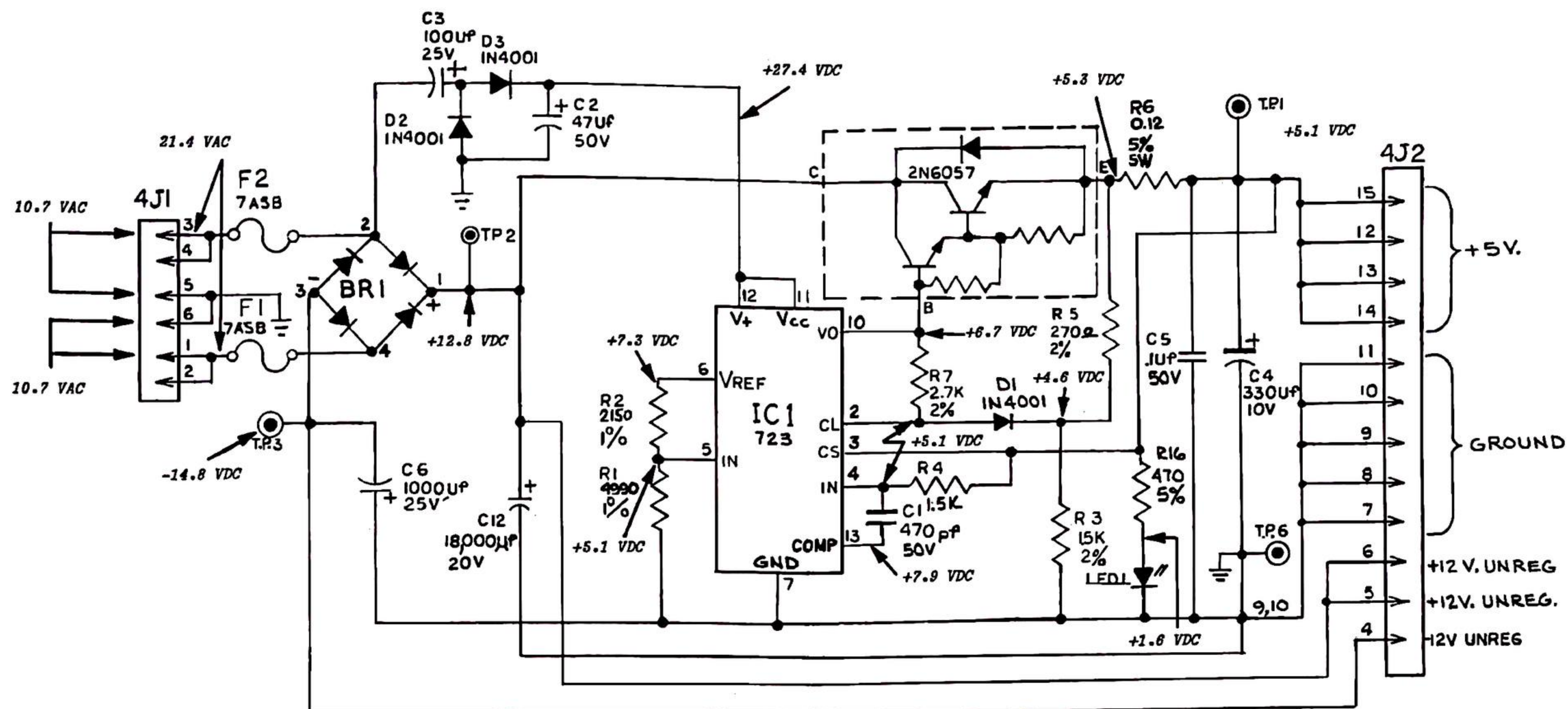
NOTES:
 1. HEAT SINK COMPOUND MUST BE APPLIED BETWEEN MICA INSULATOR AND HEAT SINK AND BETWEEN MICA INSULATOR AND TRANSISTOR.



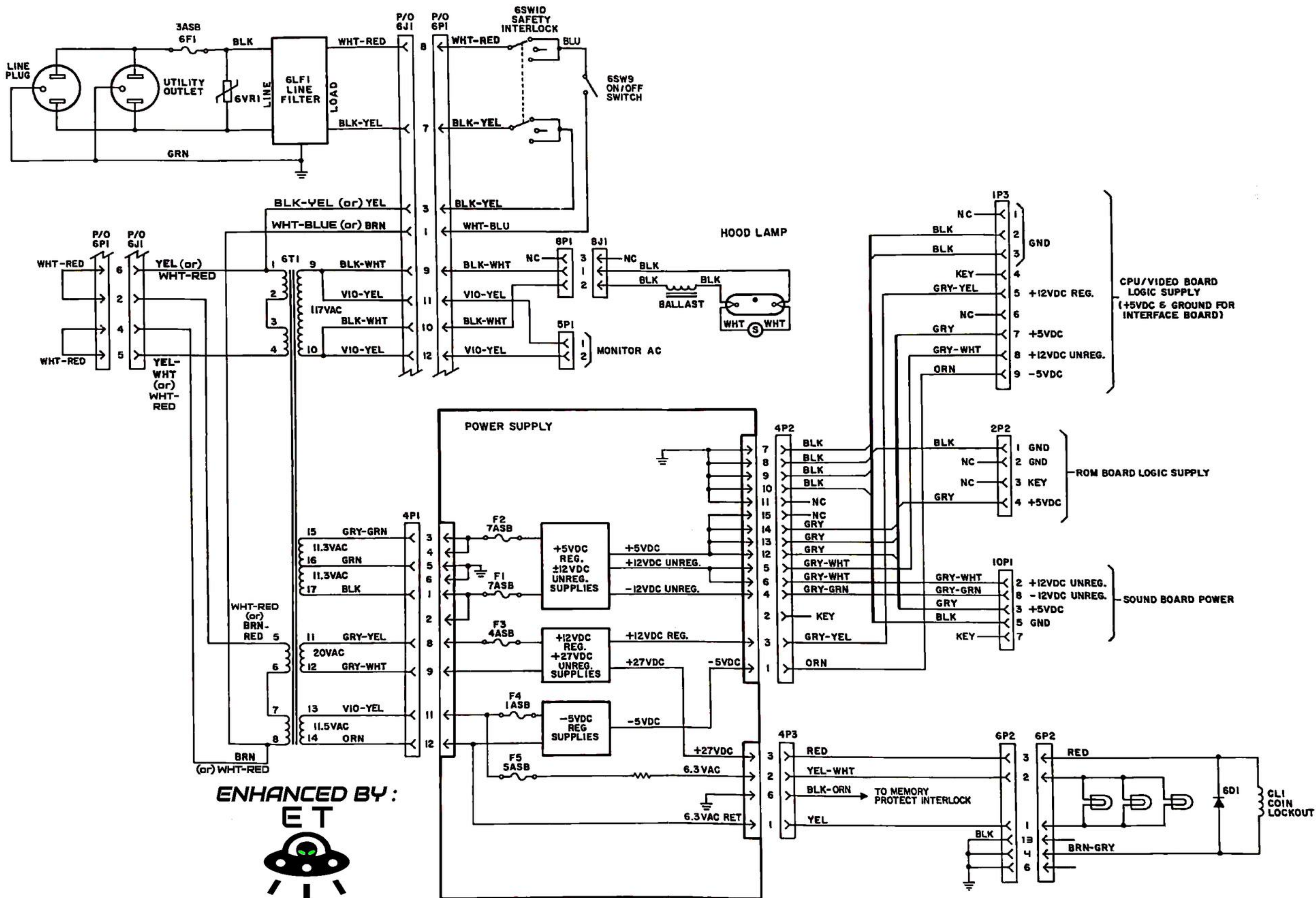
BILL OF MATERIAL				
ITEM NO	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D NO.
1	5113-09414-00	P.C. BOARD	BARE P.C. BOARD	1
2	5010-09085-00	R4	RESISTOR, 1.5K 5%, 1/4W	1
3	5010-09541-00	R8, R7	RESISTOR, 2.7K 2%, 1/4W	2
4	5010-09508-00	R5	RESISTOR, 270 2%, 1/4W	1
5	5010-09428-00	R3	RESISTOR, 1.5K 2%, 1/4W	1
6	5010-09509-00	R11	RESISTOR, 1.1K 2%, 1/4W	1
7	5010-09510-00	R13	RESISTOR, 12K 2%, 1/4W	1
8	5013-09542-00	R10	RESISTOR, 7.3K 1%, 1/4W	1
9	5013-09427-00	R1 & R9	RESISTOR, 499 1%, 1/4W	2
10	5013-09426-00	R2	RESISTOR, 2.15K 1%, 1/4W	1
11	5017-09429-00	R6	RESISTOR, 12 5%, 5W	1
12	5017-09512-00	R14	RESISTOR, 27 5%, 5W	1
13	5040-09421-00	C3 & C7	CAPACITOR, 100MFD, 25V, RADIAL	2
14	5040-09422-00	C2	CAPACITOR, 47MFD, 50V, RADIAL	1
15	5040-09420-00	C6	CAPACITOR, 1000MFD, 25V, RADIAL	1
16	5040-09413-00	C12	CAPACITOR, 1000MFD, 20V, RADIAL	1
17	5040-09423-00	C4	CAPACITOR, 330MFD, 10V, RADIAL	1
18	5040-09504-00	C13	CAPACITOR, 4700MFD, 35V, RADIAL	1
19	5043-09594-00	C5, C8, C11	CAPACITOR, 1MFD, 50V, AXIAL 85C	3
20	5043-09046-00	C1	CAPACITOR, 470 PFD, 50V, AXIAL X7R	1
21	5043-09492-00	C9	CAPACITOR, 100 PFD, 50V, AXIAL 62L	1
22	5040-09504-00	C14	CAPACITOR, 220MFD, 16V, RADIAL	1
23	5040-09493-00	C10	CAPACITOR, 22MFD, 10V, RADIAL	1
24	5070-06258-00	D1, D2, D3	DIODE, 1N4001	3
25	5100-09418-00	BR1	BRIDGE RECTIFIER, 35AMP, 100V	1
26	5100-09543-00	BR2	DIODE BRIDGE, 40A, 50V	1
27	5100-09514-00	BR3	DIODE BRIDGE, 10A, 50V	1
28	5731-09452-00	F1, F2	FUSE, 7A, 50V	2
29	5731-06314-00	F3	FUSE, 4A, 50V, 250 V	1
30	5731-06563-00	F4	FUSE, 1A, 50V	1
31	B-8416-00		HEAT SINK ASSEMBLY (SEE A)	1
32	5705-09543-00		HEAT SINK	1
33	5731-09074-00	4J2	HEADER, 15 PIN, 09-65-1151	1
34	5731-09043-00	4J1	HEADER, 12 PIN, 09-65-1121	1
35	5250-09515-00	Q2	VOLTAGE REGULATOR 7905	1
36	5440-09424-00	IC1, IC2	VOLTAGE REGULATOR 783	2
37	5824-09248-00	TP2, TP3, TP5, TP6	TERMINAL #1502-1 (TEST POST)	4
38	20-3729-00		THERMAL COMPOUND .02 OUNCE	AR
39	4005-04091-07		5-40x2/16 RH, MECH. SCREW	6
40	4403-01117-00		5-40 HEX NUT	6
41	5161-09552-00	Q1	2N3055 TRANSISTOR NPN	1
42	5731-09651-00	F5	FUSE, 5A, 50V	1
43	5731-09038-00	4J3	HEADER, 6 PIN, 09-65-1061	1
44	5012-09544-00	R15	RESISTOR, 2.0 5%, 5W	1
45	5010-09416-00	R16, R18	RESISTOR, 470 5%, 1/4W	2
46	5010-09314-00	R17	RESISTOR, 1.2K 5%, 1/4W	1
47	5704-09538-00		INSULATOR, MICA, 3mil	1
48	5671-09019-00	LED1, LED2, LED3	LED, RED	3
49	03-7558		INSULATOR, NYLON	2
50	03-7520		TIE WRAP	1
51	5732-09178-00		FUSE HOLDER	10

ENHANCED BY:

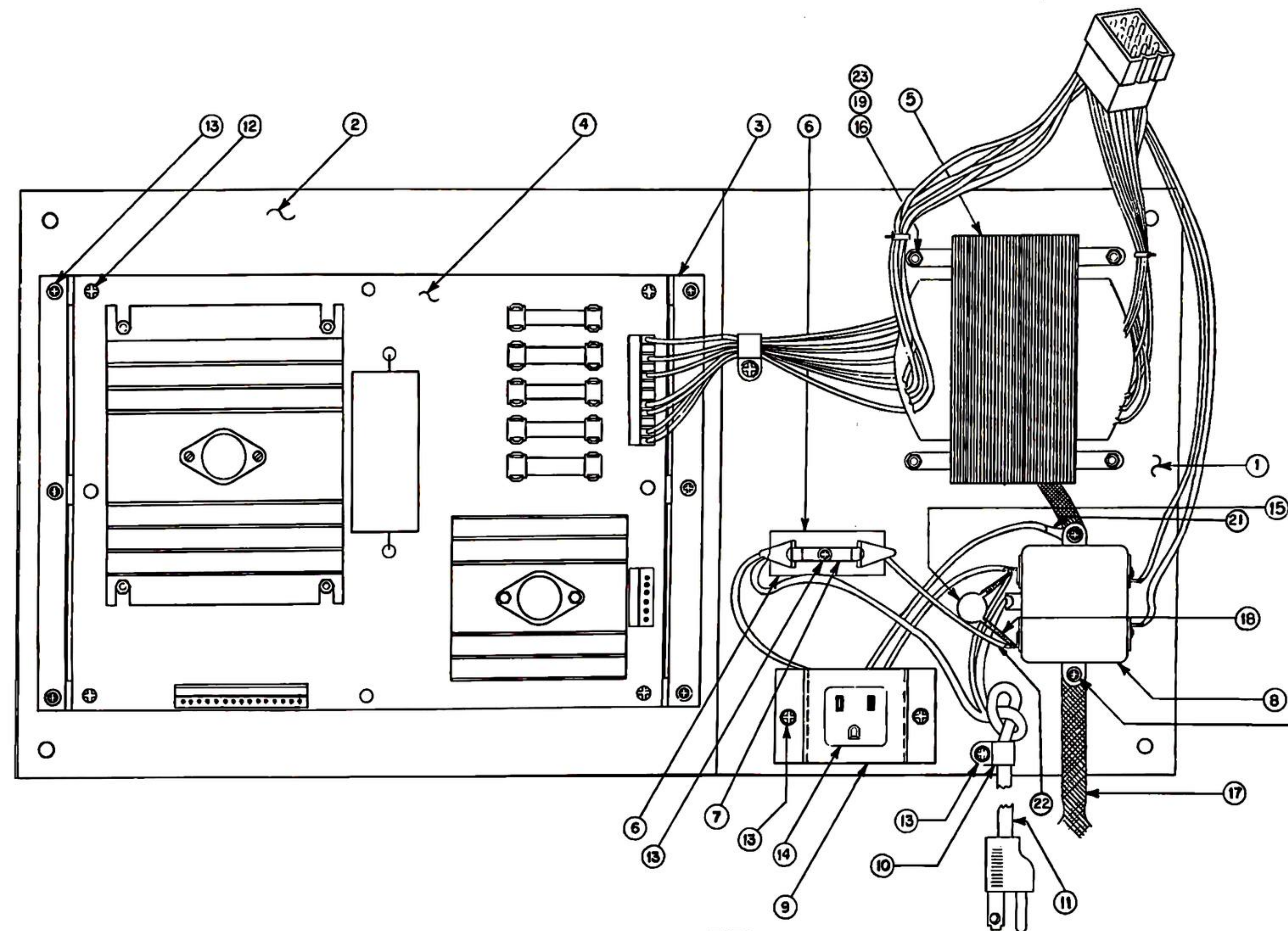




D8359 Power Supply Board
Schematic Diagram



Power Wiring Diagram



BILL OF MATERIALS			
ITEM	PART N°	DESCRIPTION	QTY.
1	11-580	POWER SUPPLY MTG. BD	1
2	01-7001	POWER SUPPLY BND PLATE	1
3	01-7002	POWER SUPPLY SUPPORT BRKT.	2
4	1D-8359	POWER SUPPLY BOARD	1
5	C85E9	VIDEO TRANSFORMER ASSY	1
6	B-6448-1	FUSE MOUNT ASS'Y.	1
7	5731-08833-00	LINE FUSE 3AMP 115V	1
8	5102-08895-00	LINE FILTER 5A	1
9	01-6948-	OUTLET MTG BRACKET	1
10	03-6048-6	CABLE CLAMP- 3/8	2
11	SEE NOTE #2	LINE CORD	1
12	4008-01018-04	MS #8-32 x 1/4 P-RH	4
13	4108-01022-08	SMS #8-32 x 1/2 P-RH-A	13
14	5851-09184-00	3WIRE SNAP-IN OUTLET	1
15	SEE NOTE #1	VARISTOR	1
16	4408-01017-00	NUT #8-32 HEX	4
17	20-9223-3	6INCH BRAIDED GROUND WIRE	1
18	20-9258	1 INCH SLEEVING INSULATION	2
19	4785-00008-00	NO#8 EXTERNAL LOCK WASHER	4
21	5825-06523-00	TERMINAL LUG	1
22	5852-09575-00	BLACK 18'AWG JUMPER WIRE-25L	1
23	4008-00003-22	MS #8-32 x 1 1/8 P-PH	4

NOTES:

1. FOR 115V.A.C., 130V. VARISTOR*5017-09044. IS USED.

FOR 230V.A.C., A 275V VARISTOR*5017-09063 IS USED.

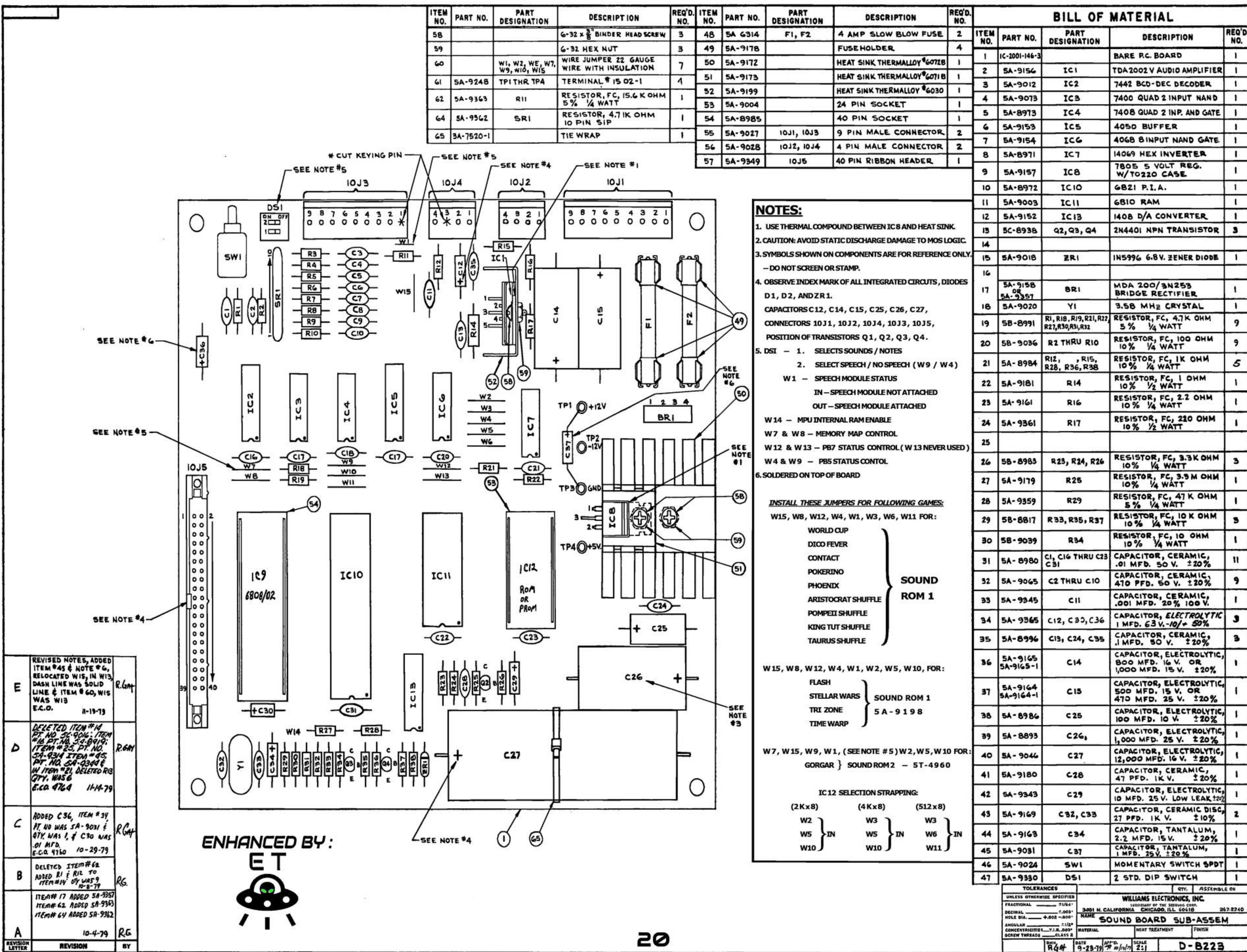
2. LINE CORD PART NUMBERS:

5850-06206-00 12 FT. EUROPE
 5850-08728-00 U.S. DOM.
 5850-08737-00 AUS.
 5850-08743-00 U.S.L.A.
 5850-09128-00 12 FT. FRANCE
 5850-09131-00 AWG 18-12 JAPAN

ENHANCED BY:

ET





ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
58			6-32 x 3/8" BINDER HEAD SCREW	3
59			6-32 HEX NUT	3
60		W1, W2, W6, W7, W9, W10, W15	WIRE JUMPER 22 GAUGE WIRE WITH INSULATION	7
61	5A-9248	TP1 THR TPA	TERMINAL #15 02-1	1
62	5A-9363	R11	RESISTOR, FC, 15.6 K OHM 5% 1/4 WATT	1
64	5A-9362	SR1	RESISTOR, 4.7 K OHM 10 PIN SIP	1
65	3A-7520-1		TIE WRAP	1

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
48	5A-6314	F1, F2	4 AMP SLOW BLOW FUSE	2
49	5A-9178		FUSE HOLDER	4
50	5A-9172		HEAT SINK THERMALLOY #6072B	1
51	5A-9173		HEAT SINK THERMALLOY #6071B	1
52	5A-9199		HEAT SINK THERMALLOY #6030	1
53	5A-9004		24 PIN SOCKET	1
54	5A-8985		40 PIN SOCKET	1
55	5A-9027	10J1, 10J3	9 PIN MALE CONNECTOR	2
56	5A-9028	10J2, 10J4	4 PIN MALE CONNECTOR	2
57	5A-9349	10J5	40 PIN RIBBON HEADER	1

BILL OF MATERIAL				
ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
1	IC-2001-146-3		BARE P.C. BOARD	1
2	5A-9156	IC1	TDA2002 V AUDIO AMPLIFIER	1
3	5A-9012	IC2	7442 BCD-DEC DECODER	1
4	5A-9073	IC3	7400 QUAD 2 INPUT NAND	1
5	5A-8973	IC4	7408 QUAD 2 INP. AND GATE	1
6	5A-9153	IC5	4050 BUFFER	1
7	5A-9154	IC6	4068 8 INPUT NAND GATE	1
8	5A-8971	IC7	14069 HEX INVERTER	1
9	5A-9157	IC8	7805 5 VOLT REG. W/TO220 CASE	1
10	5A-8972	IC10	6821 P.I.A.	1
11	5A-9003	IC11	6810 RAM	1
12	5A-9152	IC13	1408 D/A CONVERTER	1
13	5C-8938	Q2, Q3, Q4	2N4401 NPN TRANSISTOR	3
14				
15	5A-9018	ZR1	1N5996 6.8V. ZENER DIODE	1
16				
17	5A-9158 OR 5A-9357	BR1	MDA 200/3N253 BRIDGE RECTIFIER	1
18	5A-9020	Y1	3.58 MHZ CRYSTAL	1
19	5B-8991	R1, R18, R19, R21, R22, R27, R30, R31, R32	RESISTOR, FC, 4.7 K OHM 5% 1/4 WATT	9
20	5B-9036	R2 THRU R10	RESISTOR, FC, 100 OHM 10% 1/4 WATT	9
21	5A-8984	R12, R15, R28, R36, R38	RESISTOR, FC, 1 K OHM 10% 1/4 WATT	5
22	5A-9181	R14	RESISTOR, FC, 1 OHM 10% 1/2 WATT	1
23	5A-9161	R16	RESISTOR, FC, 2.2 OHM 10% 1/4 WATT	1
24	5A-9361	R17	RESISTOR, FC, 220 OHM 10% 1/2 WATT	1
25				
26	5B-8983	R23, R24, R26	RESISTOR, FC, 3.3 K OHM 10% 1/4 WATT	3
27	5A-9179	R25	RESISTOR, FC, 3.3 M OHM 10% 1/4 WATT	1
28	5A-9359	R29	RESISTOR, FC, 47 K OHM 5% 1/4 WATT	1
29	5B-8817	R33, R35, R37	RESISTOR, FC, 10 K OHM 10% 1/4 WATT	3
30	5B-9039	R34	RESISTOR, FC, 10 OHM 10% 1/4 WATT	1
31	5A-8980	C1, C16 THRU C23, C31	CAPACITOR, CERAMIC, .01 MFD. 50 V. ±20%	11
32	5A-9065	C2 THRU C10	CAPACITOR, CERAMIC, 470 PFD. 50 V. ±20%	9
33	5A-9345	C11	CAPACITOR, CERAMIC, .001 MFD. 20% 100 V.	1
34	5A-9365	C12, C30, C36	CAPACITOR, ELECTROLYTIC 1 MFD. 63 V. ±50%	3
35	5A-8996	C13, C24, C35	CAPACITOR, CERAMIC, .1 MFD. 50 V. ±20%	3
36	5A-9165 5A-9165-1	C14	CAPACITOR, ELECTROLYTIC, 800 MFD. 16 V. OR 1,000 MFD. 15 V. ±20%	1
37	5A-9164 5A-9164-1	C15	CAPACITOR, ELECTROLYTIC, 500 MFD. 15 V. OR 470 MFD. 25 V. ±20%	1
38	5A-8986	C25	CAPACITOR, ELECTROLYTIC, 100 MFD. 10 V. ±20%	1
39	5A-8893	C26	CAPACITOR, ELECTROLYTIC, 1,000 MFD. 25 V. ±20%	1
40	5A-9046	C27	CAPACITOR, ELECTROLYTIC, 12,000 MFD. 16 V. ±20%	1
41	5A-9180	C28	CAPACITOR, CERAMIC, 47 PFD. 1 K V. ±20%	1
42	5A-9343	C29	CAPACITOR, ELECTROLYTIC, 10 MFD. 25 V. LOW LEAK ±20%	1
43	5A-9169	C32, C33	CAPACITOR, CERAMIC DISC, 27 PFD. 1 K V. ±10%	2
44	5A-9163	C34	CAPACITOR, TANTALUM, 2.2 MFD. 15 V. ±20%	1
45	5A-9031	C37	CAPACITOR, TANTALUM, 1 MFD. 25 V. ±20%	1
46	5A-9024	SW1	MOMENTARY SWITCH SPDT	1
47	5A-9330	DS1	2 STD. DIP SWITCH	1

NOTES:

- USE THERMAL COMPOUND BETWEEN IC8 AND HEAT SINK.
- CAUTION: AVOID STATIC DISCHARGE DAMAGE TO MOS LOGIC.
- SYMBOLS SHOWN ON COMPONENTS ARE FOR REFERENCE ONLY. - DO NOT SCREEN OR STAMP.
- OBSERVE INDEX MARK OF ALL INTEGRATED CIRCUITS, DIODES D1, D2, AND ZR1.
CAPACITORS C12, C14, C15, C25, C26, C27, CONNECTORS 10J1, 10J2, 10J4, 10J3, 10J5, POSITION OF TRANSISTORS Q1, Q2, Q3, Q4.
- DSI - 1. SELECTS SOUNDS / NOTES
2. SELECT SPEECH / NO SPEECH (W9 / W4)
W1 - SPEECH MODULE STATUS
IN - SPEECH MODULE NOT ATTACHED
OUT - SPEECH MODULE ATTACHED
W14 - MPU INTERNAL RAM ENABLE
W7 & W8 - MEMORY MAP CONTROL
W12 & W13 - PB7 STATUS CONTROL (W13 NEVER USED)
W4 & W9 - PB5 STATUS CONTROL
- SOLDERED ON TOP OF BOARD

INSTALL THESE JUMPERS FOR FOLLOWING GAMES:

W15, W8, W12, W4, W1, W3, W6, W11 FOR:

WORLD CUP	SOUND ROM 1
DICO FEVER	
CONTACT	
POKERINO	
PHOENIX	
ARISTOCRAT SHUFFLE	
POMPEII SHUFFLE	
KING TUT SHUFFLE	
TAURUS SHUFFLE	

W15, W8, W12, W4, W1, W2, W5, W10, FOR:

FLASH	SOUND ROM 1
STELLAR WARS	
TRI ZONE	
TIME WARP	

W7, W15, W9, W1, (SEE NOTE #5) W2, W5, W10 FOR:

GORGAR } SOUND ROM2 - 5T-4960

IC12 SELECTION STRAPPING:

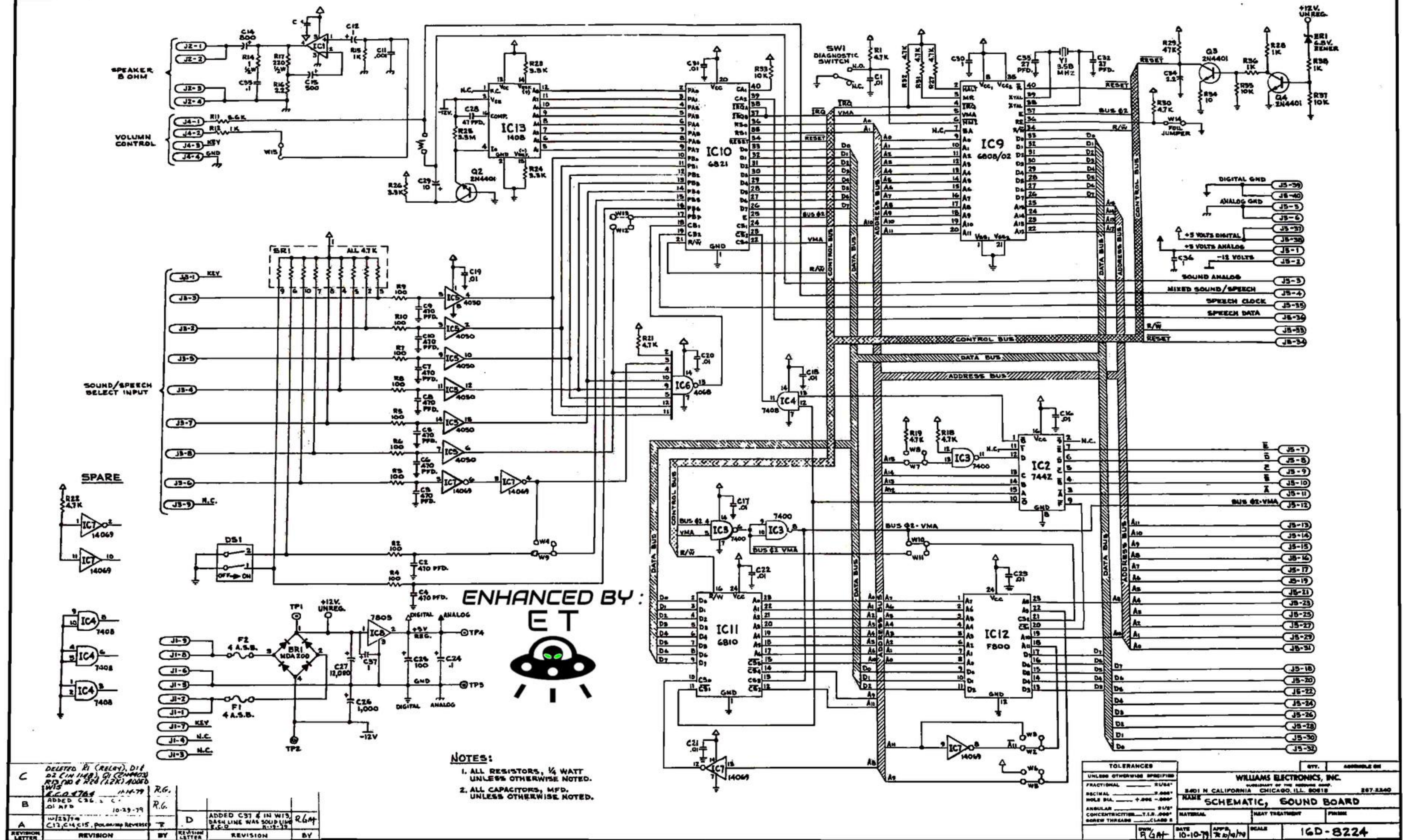
(2Kx8)	(4Kx8)	(512x8)
W2 } IN	W3 } IN	W3 } IN
W5 } IN	W5 } IN	W6 } IN
W10 } IN	W10 } IN	W11 } IN

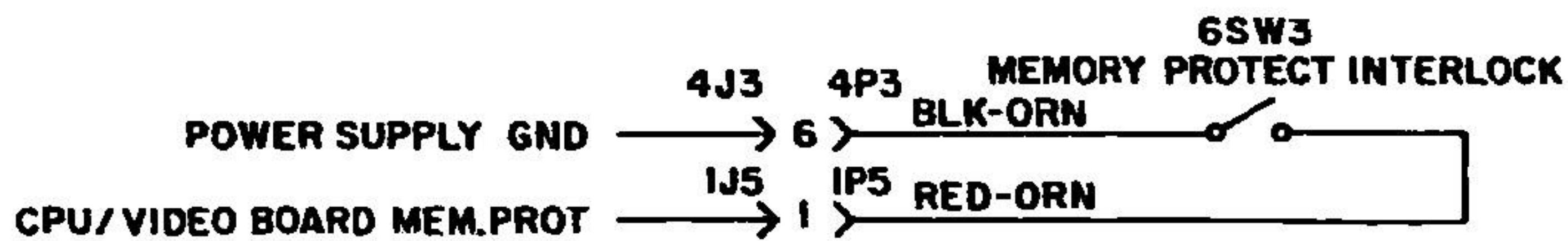
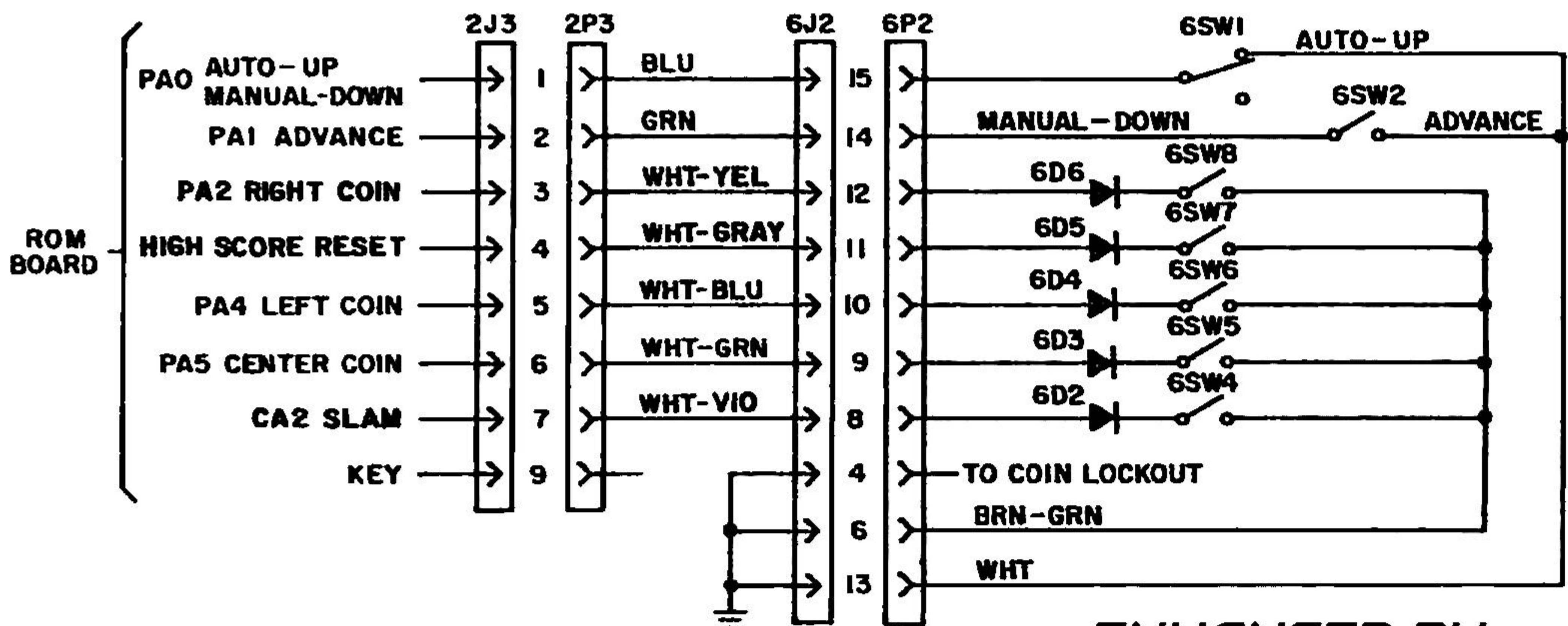
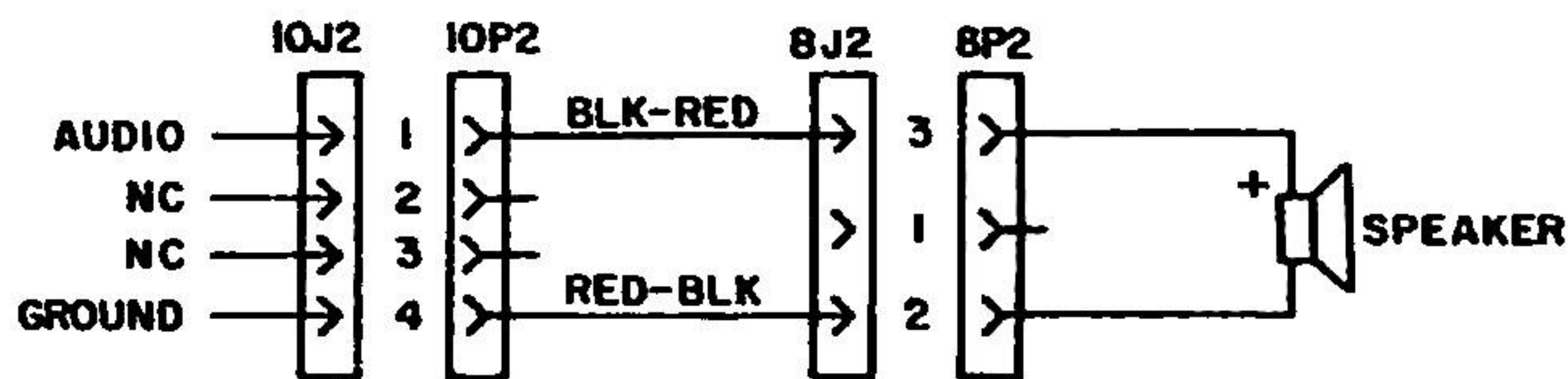
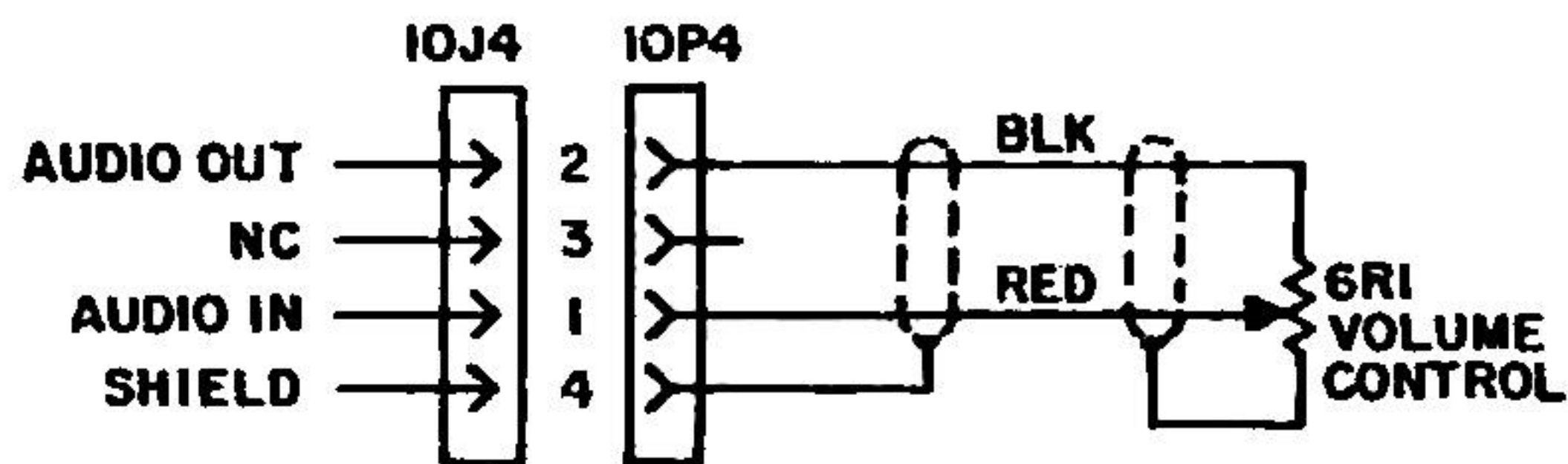
REVISION LETTER	REVISION	BY
E	REVISED NOTES, ADDED ITEM #45 & NOTE #6, RELOCATED W15, IN W13, DASH LINE WAS SOLID LINE & ITEM #60, W15 WAS W13 E.C.O. 11-19-79	R.G.M.
D	DELETED ITEM #14 PT. NO. 5A-9016; ITEM #16 PT. NO. 5A-8919; ITEM #25, PT. NO. 5A-9314 ITEM #45, PT. NO. 5A-9344 & IN ITEM #21, DELETED R13 QTY. WAS 6 E.C.O. 4764 11-14-79	R.G.M.
C	ADDED C36, ITEM #34, PT. NO. 5A-9031 & QTY. WAS 1, & C30 WAS .01 MFD. E.C.O. 4760 10-29-79	R.G.M.
B	DELETED ITEM #62 ADDED R1 & R12 TO ITEM #14 QTY. WAS 9 10-8-79	R.G.
A	ITEM #17 ADDED 5A-9357 ITEM #62 ADDED 5A-9363 ITEM #64 ADDED 5A-9362 10-4-79	R.G.



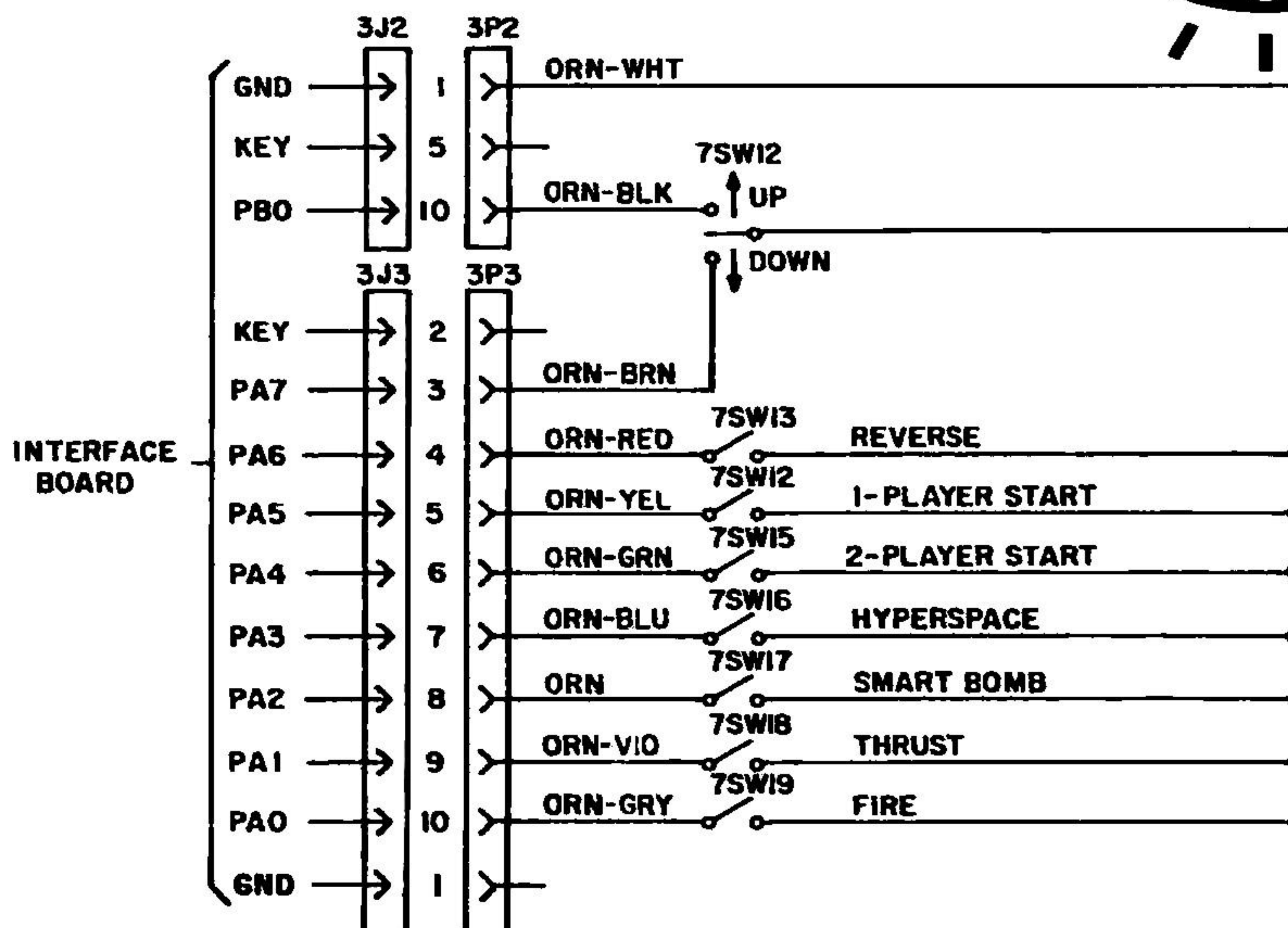
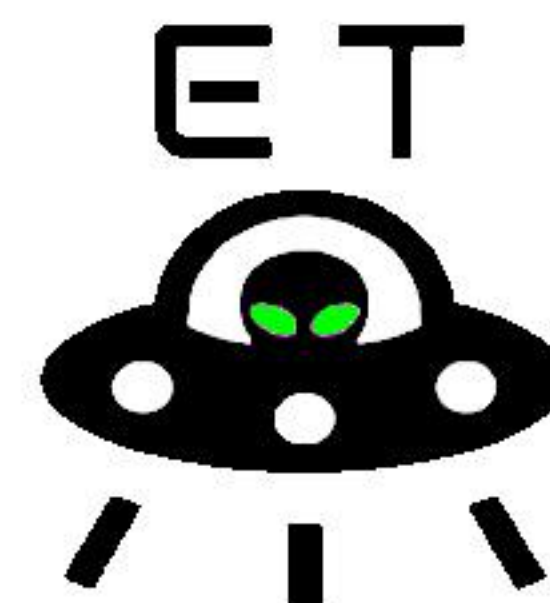
D8224 Sound Board Assembly Drawing
(LATER SERIES BOARD)

TOLERANCES		QTY.		ASSEMBLE ON	
UNLESS OTHERWISE SPECIFIED					
FRACTIONAL	±.004"				
DECIMAL	±.001"				
HOLE DIA.	+.002" - .005"				
ANGULAR	±1/2°				
CONCENTRICITY	±.005"				
SCREW THREADS	CLASS 2				
WILLIAMS ELECTRONICS, INC.					
3001 N. CALIFORNIA CHICAGO, ILL. 60618 267-2240					
NAME: SOUND BOARD SUB-ASSEM					
MATERIAL		HEAT TREATMENT		FINISH	
DATE: 9-23-79		APPROVED: [Signature]		SCALE: 2:1	
DWG: R44		D-B223			



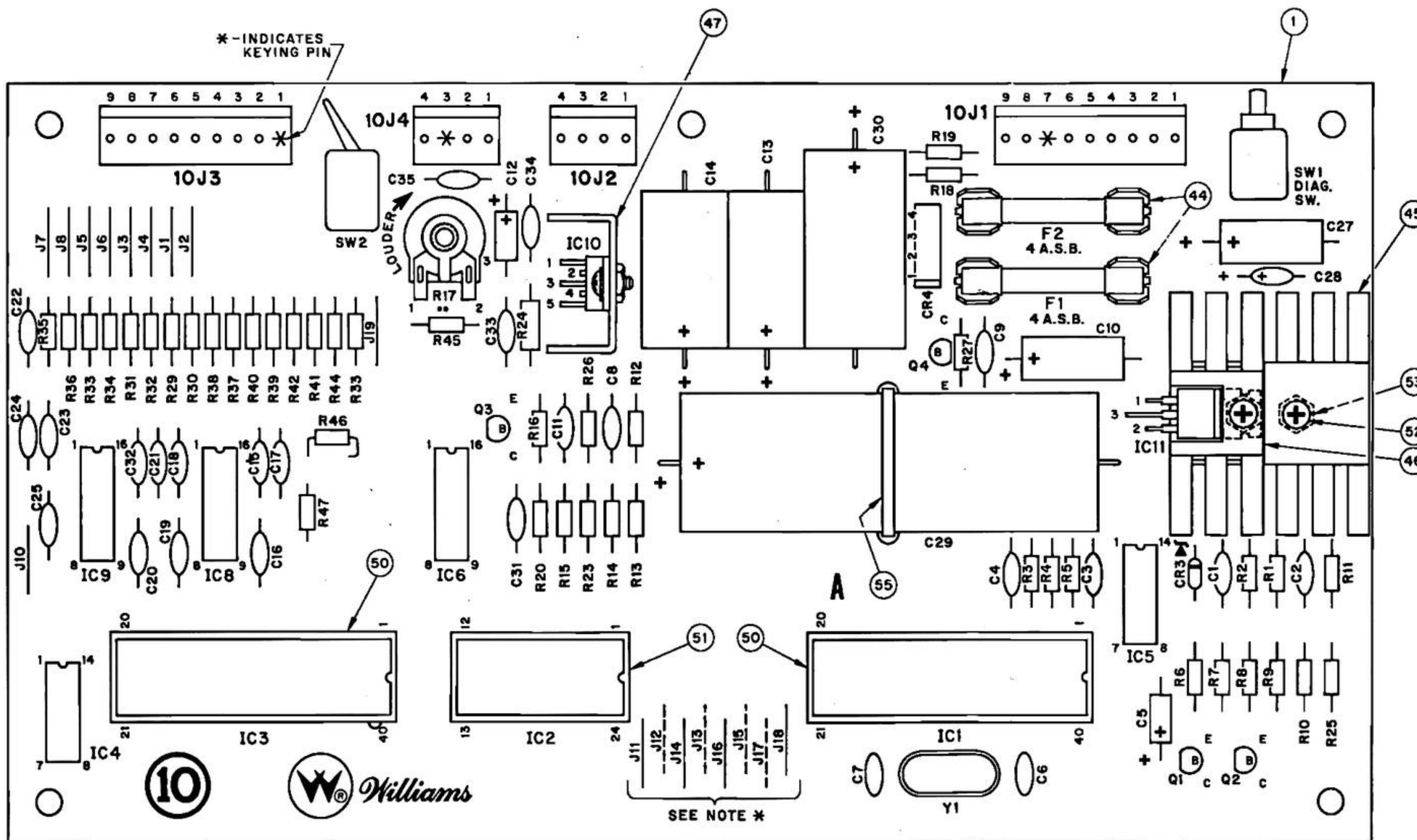


ENHANCED BY :



CABINET WIRING DIAGRAM

ENHANCED BY:
ET

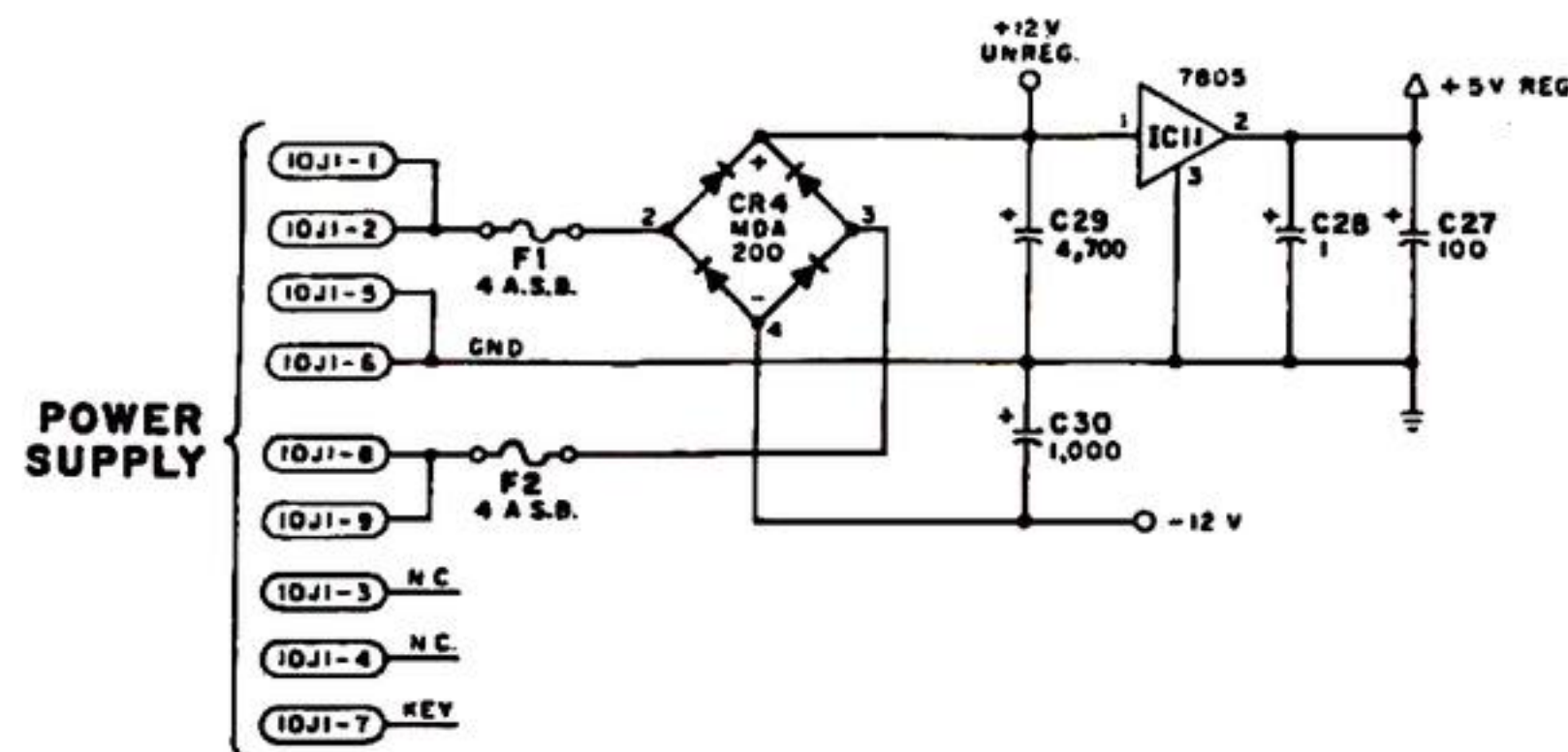


* NOTE:
JUMPERS SHOWN WITH SOLID LINES CONNECTED FOR
PROM OPERATION. THOSE SHOWN WITH DASHED LINES
CONNECTED FOR ROM OPERATION.

BILL OF MATERIAL

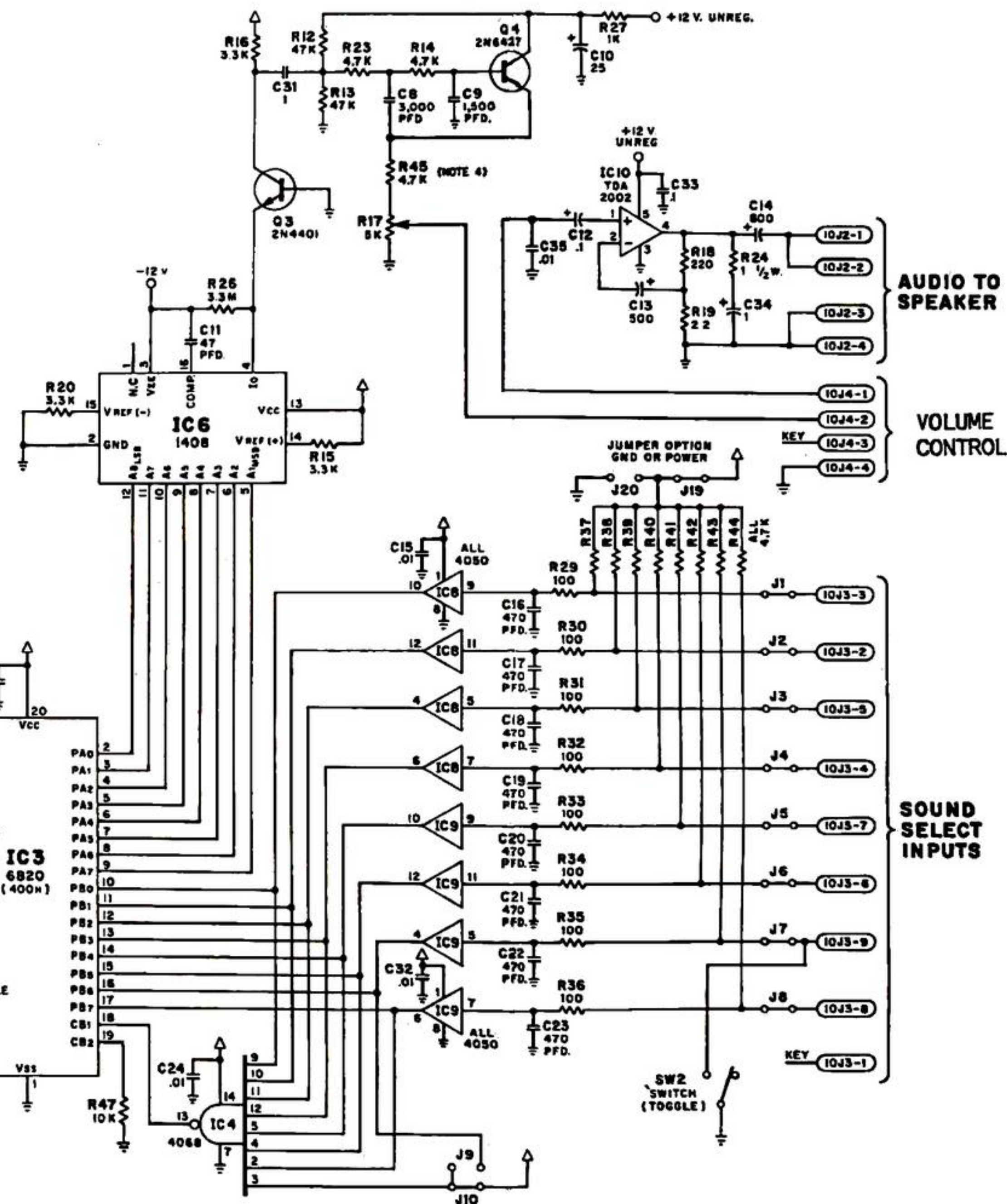
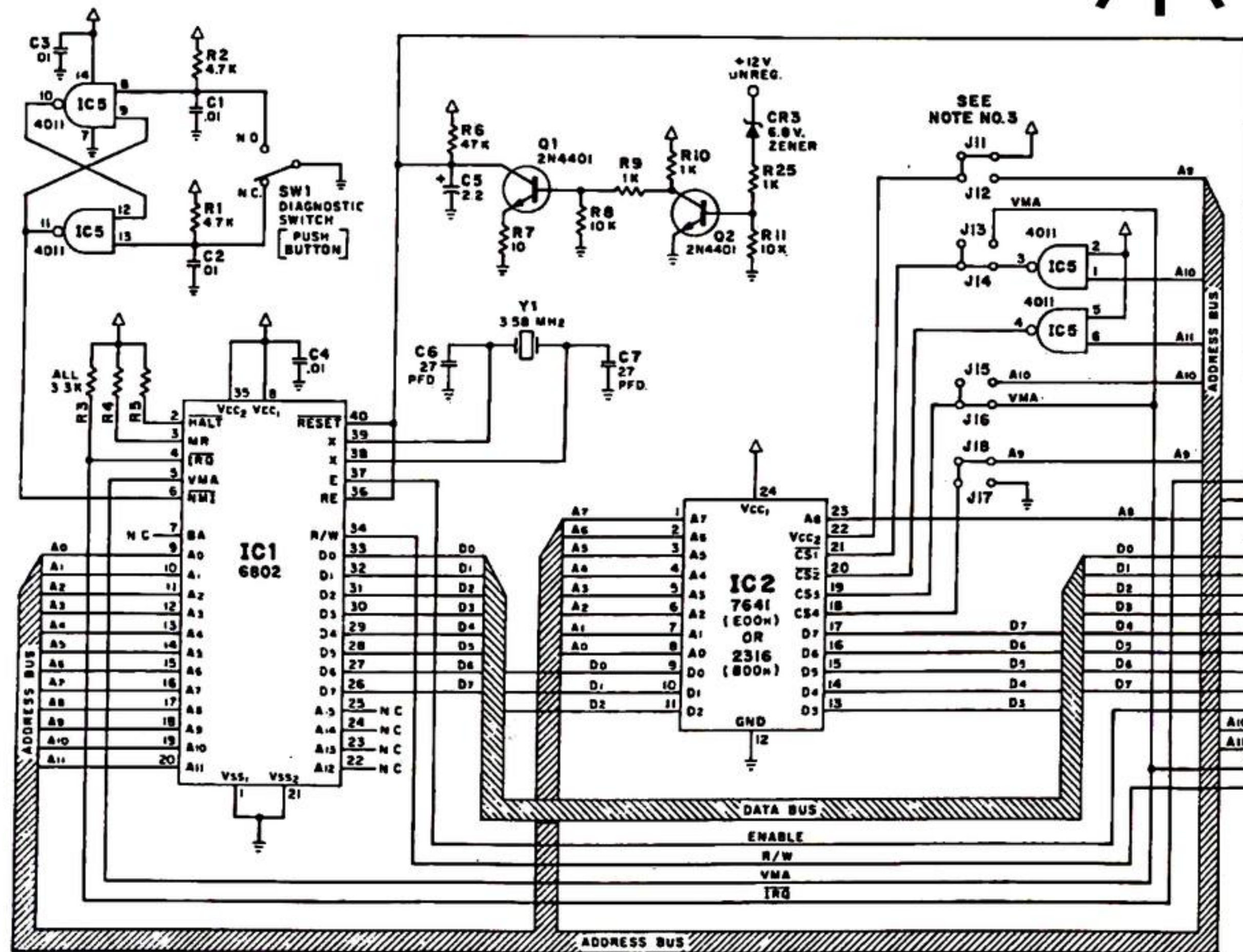
ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
1	1B-2001-137		BARE P.C. BOARD	1
2	5A-8972	IC3	MC6820 PIA	1
3	5A-9154	IC4	4068 8 INPUT NAND GATE	1
4	5A-9155	IC5	4011 QUAD 2 INPUT NAND GATE	1
5	5A-9152	IC6	MC1408 D/A CONVERTER	1
6	5A-9153	IC8, IC9	4050 BUFFER	2
7	5A-9156	IC10	YDA2002V AUDIO AMPLIFIER	1
8	5A-9157	IC11	7805 5 VOLT REG. W/TO220 CASE	1
9	5C-8938	Q1, Q2, Q3	2N4401 NPN TRANSISTOR	3
10	5A-8976	Q4	2N6427 NPN TRANSISTOR	1
11	5A-9018	CR3	1N5996 6.8 V ZENER DIODE	1
12	5A-9158	CR4	MDA200/3N253 BRIDGE RECTIFIER	1
13	5B-8991	R1, R2, R14, R23, R37 THRU R45	RESISTOR, FC, 4.7K OHM 10% 1/4 W	13
14	5B-8983	R3, R4, R5, R15, R16, R20	RESISTOR, FC, 3.3 K OHM 10% 1/4 W	6
15	5A-8984	R9, R10, R25, R27	RESISTOR, FC, 1K OHM 10% 1/4 W	5
16	5B-9039	R7	RESISTOR, FC, 10 OHM 10% 1/4 W	1
17	5B-8817	R8, R11, R46, R47	RESISTOR, FC, 10K OHM 10% 1/4 W	4
18	5B-9035	R6, R12, R13	RESISTOR, FC, 47 K OHM 10% 1/4 W	3
19	5A-9160	R18	RESISTOR, FC, 220 OHM 10% 1/4 W	1
20	5A-9161	R19	RESISTOR, FC, 2.2 OHM 10% 1/4 W	1
21	5A-9179	R26	RESISTOR, FC, 3.3M OHM 10% 1/4 W	1
22	5B-9036	R29 THRU R36	RESISTOR, FC, 100 OHM 10% 1/4 W	8
23	5A-9181	R24	RESISTOR, FC, 1 OHM 10% 1/2 W	1
24	5A-9376	C29	CAPACITOR, ELECTROLYTIC, 14,700 MFD. 16 V.	1
25	5A-8986	C27	CAPACITOR, ELECTROLYTIC, 100 MFD. 10 V.	1
26	5A-8893	C30	CAPACITOR, ELECTROLYTIC, 1,000 MFD. 25 V.	1
27	5A-9164 5A-9164-1	C13	CAPACITOR, ELECTROLYTIC, 500 MFD. 15 V. OR 470 MFD. 25 V.	1
28	5A-9165 5A-9165-1	C14	CAPACITOR, ELECTROLYTIC, 800 MFD. 16 V. OR 1,000 MFD. 15 V.	1
29	5A-9168 5A-9168-1	C10	CAPACITOR, ELECTROLYTIC, 25 MFD. 20 V. OR 33 MFD. 16 V.	1
30	5A-9031	C12	CAPACITOR, TANTALUM, 1 MFD. 25 V.	1
31	5A-9163	C5	CAPACITOR, TANTALUM, 2.2 MFD. 15 V.	1
32	5A-8980	C1 THRU C4, C15, C24, C25, C32, C35	CAPACITOR, CERAMIC, .01 MFD. 50 V.	9
33	5A-8996	C28, C31, C33, C34	CAPACITOR, CERAMIC, .1 MFD. 50 V.	4
34	5A-9169	C6, C7	CAPACITOR, CERAMIC DISC, 27 PFD. 1K V.	2
35	5A-9063	C16 THRU C23	CAPACITOR, CERAMIC DISC, 470 PFD. 50 V.	8
36	5A-9166	C9	CAPACITOR, CERAMIC DISC, .0015 MFD. 1K V.	1
37	5A-9167	C8	CAPACITOR, CERAMIC DISC, .003 MFD. 1K V.	1
38	5A-9180	C11	CAPACITOR, CERAMIC DISC, 47 PFD. 1K V.	1
39	5A-9185	R17	POTENTIOMETER, 5K OHM	1
40	5A-9020	Y1	CRYSTAL, 3.58 MHE.	1
41	5A-9024	SW1	MOMENTARY SWITCH SPDT	1
42	5A-9171	SW2	TOGGLE SWITCH SPDT	1
43	5A-6314	F1, F2	4 AMP. SLOW BLOW FUSE	2
44	5A-9178		FUSEHOLDER	4
45	5A-9172		HEAT SINK THERMALLOY #6072B	1
46	5A-9173		HEAT SINK THERMALLOY #6071B	1
47	5A-9199		HEAT SINK THERMALLOY #8030	1
48	5A-9027	10J1, 10J3	9 PIN MALE CONNECTOR	2
49	5A-9028	10J2, 10J4	4 PIN MALE CONNECTOR	2
50	5A-8985		40 PIN SOCKET	2
51	5A-9004		24 PIN SOCKET	1
52			6-32 X 3/8 BINDER HEAD SCREW	2
53			6-32 HEX. NUT	2
54		J1 THRU J8, J10, THRU J19	WIRE JUMPER 22 GAUGE WIRE WITH INSULATION	18
55	5A-7520-1		TIE WRAP	1
56	5A-9150	IC1	MC6802 MICROPROCESSOR	1
57	5A-9264	IC2	VIDEO SOUND ROM 1	1

*ON MOST BOARDS, R17 IS REMOVED AND A JUMPER
CONNECTED BETWEEN TERMINALS 1 AND 2.



- NOTES:**
1. UNLESS OTHERWISE INDICATED ALL RESISTORS IN OHMS, 1/4 WATT.
 2. UNLESS OTHERWISE INDICATED ALL CAPACITORS ARE IN MICROFARADS.
 3. J11, J14, J16, AND J18 ARE REMOVED AND J12, J13, J15, AND J17 ARE CONNECTED.
 4. ON MOST BOARDS R17 IS REMOVED AND R45 IS CONNECTED THROUGH A JUMPER TO 10J4-2.

ENHANCED BY:



PARTS LIST
ARRANGED BY GROUPS OF RELATED PARTS

DRAWING DESIGNATION	PART NUMBER	PINOUT #	DESCRIPTION
	D-8356 5770-09471-00		CPU/VIDEO Board Assembly Bare P.C. Board
INTEGRATED CIRCUITS			
ØD	5281-09532-00	16	74LS133 13-Input NAND
1C, 1D, 1F, 2K	5370-08989-00 or	16	8T97 Hex, TS Buffer
	5281-09385-00	16	74LS367 Hex, TS Buffer
1G	5342-09015-00	24	7641 512 x 8 PROM (Horiz. Cntrl.)
1I	5340-09017-00	22	5101 256 x 4 CMOS RAM
1J, 5E	5280-09013-00	14	7404 Hex Inverter
1K	5310-09237-00	14	4071 CMOS Quad 2-Input OR
1L, 10	5340-09625-00	16	7489 16 x 4 RAM
2D	5400-09490-00	40	6809E Microprocessor
2F	5281-09246-00	16	74LS139 Dual 2 to 4 Line Decoder
2H	5280-08973-00	14	7408 Quad AND
2I, 6G, 6H	5281-09485-00	16	74LS257 Quad Data Multiplexer
2L, 2N, 20, 2Q	5280-09489-00	16	9316 4-Bit Bin. Counter
2R, 5F	5280-09475-00	14	7411 Triple 3-Input AND
2S	5280-09073-00	14	7400 Quad 2-Input NAND
3B	5281-09215-00	14	74LS04 Hex Inverter
3C	5281-09533-00	14	74LS393 Dual 4-Bit Bin. Counter
3D, 3G, 4G	5280-08948-00	14	7402 Quad 2-Input NOR
3F	5281-09308-00	20	74LS245 Octal Bus Transceiver
3H, 4F	5280-09547-00	14	7427 Triple 3-Input POS-NOR
3I, 4I, 5I, 6I	5280-09482-00	16	74165 8-Bit Shift Register
3K, 4B, 5D, 5H, 6D, 6E	5280-09478-00	14	7474 Dual D-Type Latch
3L, 3N, 30, 3Q	5280-09481-00	16	74153 Dual 4 to 1 Data Multiplexer
3R	5280-09476-00	14	7421 Dual 4-Input AND
3S	5280-09530-00	14	7420 Dual 4-Input NAND
4C	5281-09480-00	14	74LS107 Dual J-K Flip-Flop
4D	5280-09479-00	14	7486 Quad 2-Input EX-OR
4E	5280-09477-00	14	7432 Quad 2-Input OR
4K, 5K, 6K	5281-09486-00	20	74LS374 Octal D-Type Flip-Flop
4L, 4M, 4N, 40, 4P, 4Q, 4R, 4S, 5L, 5M, 5N, 5O, 5P, 5Q, 5R, 5S, 6L, 6M, 6N, 6O, 6P, 6Q, 6R, 6S	5340-09488-00	16	4116 16K x 1 Dynamic RAM
5G, 6F	5280-09011-00	14	7410 Triple 3-Input NAND
RESISTORS			
R1	5010-09039-00		Resistor, FC, 10 Ohm, 5%, 1/4W
R2, 5, 6, 64	5010-09241-00		Resistor, FC, 22K Ohm, 5%, 1/4W
R3	5010-09113-00		Resistor, FC, 33K Ohm, 5%, 1/4W
R4, 7	5010-08997-00		Resistor, FC, 2.7K Ohm, 5%, 1/4W
R8	5010-09359-00		Resistor, FC, 47K Ohm, 5%, 1/4W
R9, 31, 45, 51, 54, 55	5010-08983-00		Resistor, FC, 3.3K Ohm, 5%, 1/4W
R10	5010-09036-00		Resistor, FC, 100 Ohm, 5%, 1/4W
R11, 21, 22, 25, 27, 28, 29, 30, 32, 42, 43, 44, 49, 52, 56, 57, 63, 67, 68	5010-08991-00		Resistor, FC, 4.7K Ohm, 5%, 1/4W

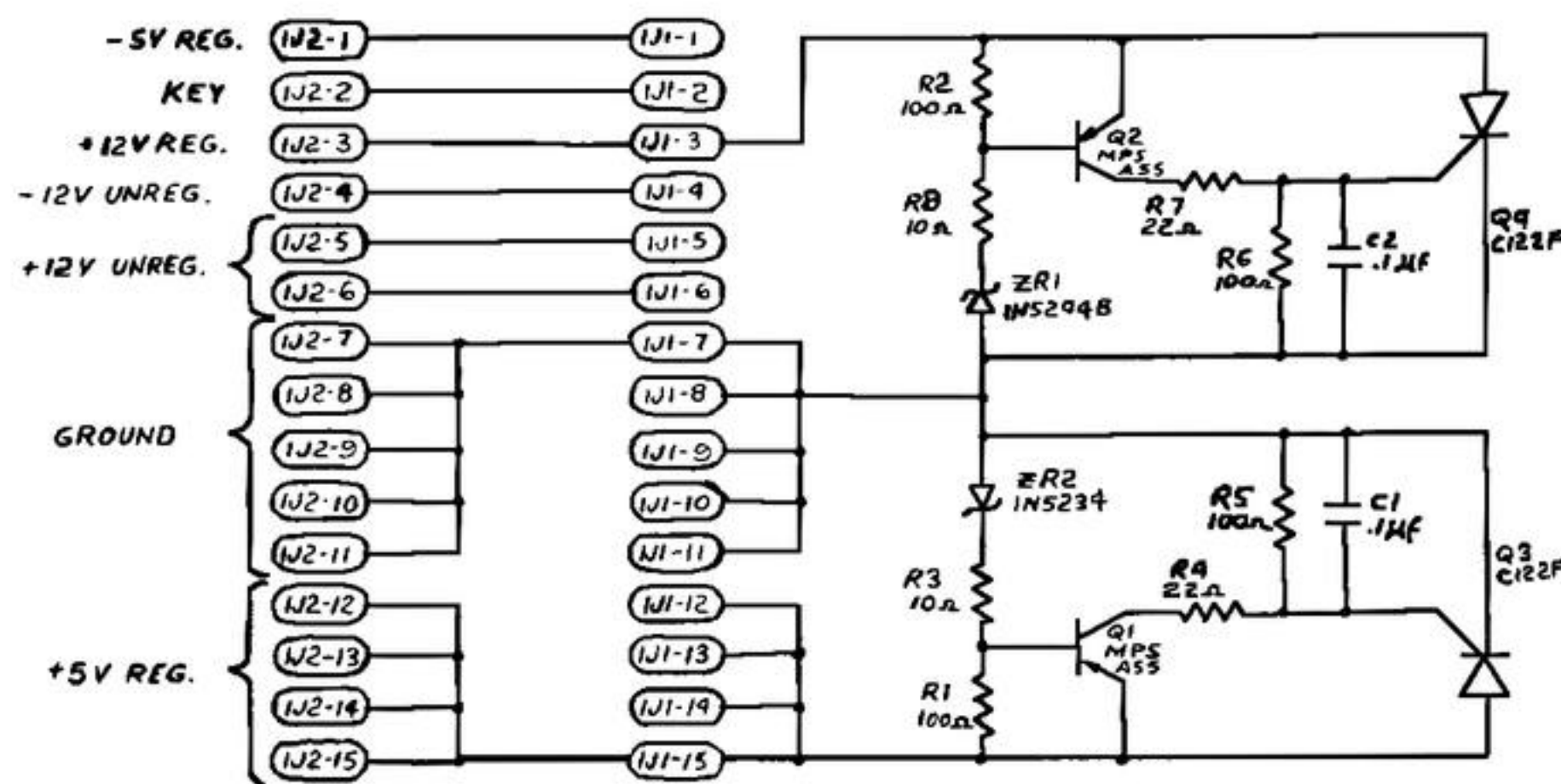
PIN #	SLOT #
14	27
16	47
20	4
22	1
24	1
40	1

PARTS LIST
ARRANGED BY GROUPS OF RELATED PARTS

DRAWING DESIGNATION	PART NUMBER	DESCRIPTION
RESISTORS (Con't.)		
R12	5010-09442-00	Resistor, FC, 330K Ohm, 5%, 1/4W
R13, 16, 18	5010-09001-00	Resistor, FC, 330 Ohm, 5%, 1/4W
R14, 15, 20	5010-08992-00	Resistor, FC, 560 Ohm, 5%, 1/4W
R17, 19	5010-09314-00	Resistor, FC, 1.2K Ohm, 5%, 1/4W
R23, 47, 48	5010-08998-00	Resistor, FC, 2.2K Ohm, 5%, 1/4W
R26, 66	5010-09034-00	Resistor, FC, 10K Ohm, 5%, 1/4W
R33, 34, 37	5010-09548-00	Resistor, FC, 68 Ohm, 2%, 1/4W
R35, 36, 38, 39,	5010-09187-00	Resistor, FC, 150 Ohm, 5%, 1/4W
R40, 41		
R46, 58, 59, 60, 61, 65	5010-09358-00	Resistor, FC, 1K Ohm, 5%, 1/4W
R62	5010-09416-00	Resistor, FC, 470 Ohm, 5%, 1/4W
CAPACITORS		
C1	5040-09545-00	Capacitor, 0.22 MFD, 10V, Axial
C2	5043-09492-00	Capacitor, 100 PFD, 50V, Axial
C3	5043-09491-00	Capacitor, 22 PFD, 1K V, Axial
C4, 5, 6	5040-09421-00	Capacitor, 100 MFD, 25V, Radial
C7 thru C11, C13 thru C40, C13 thru C40, C50 thru C89	5043-08996-00	Capacitor, 0.1 MFD, 50V, Axial
C12	5043-09065-00	Capacitor, 470 PFD, 50V, Axial
C41 thru C49 not used		
CONNECTORS		
1J1	5791-09349-00	40 Pin Ribbon Cable Header
1J2	5791-09437-00	20 Pin Ribbon Cable Header
1J3	5791-09027-00	9 Pin Header
1J4	5791-09497-00	7 Pin Header
1J5	5791-09028-00	4 Pin Header
MISCELLANEOUS		
ZR1	5075-09059-00	1N5990A Zener Diode, 3.9V
ZR2	5075-09018-00	1N5996A Zener Diode, 6.8V
D1 not used		
D2 thru D10, D12 thru D15	5070-08919-00	1N4148 Diode
D11	5070-09266-00	1N5817 Diode
Q1 thru Q4, Q6, 11, 12	5160-08938-00	2N4401 Transistor
Q5, Q7 thru Q10	5190-09016-00	2N4403 Transistor
CR1	5520-09495-00	Crystal, 12 MHz
SW1	5641-09371-00	Pushbutton Sw. (Momentary ON)
B1, 2, 3	5880-09022-00	AA Batteries, Alkaline, 1.5V
--	5881-09021-00	Battery Holder #171
--	5700-09006-00	16-Pin IC Socket
--	5700-09004-00	24-Pin IC Socket
--	5700-08985-00	40-Pin IC Socket

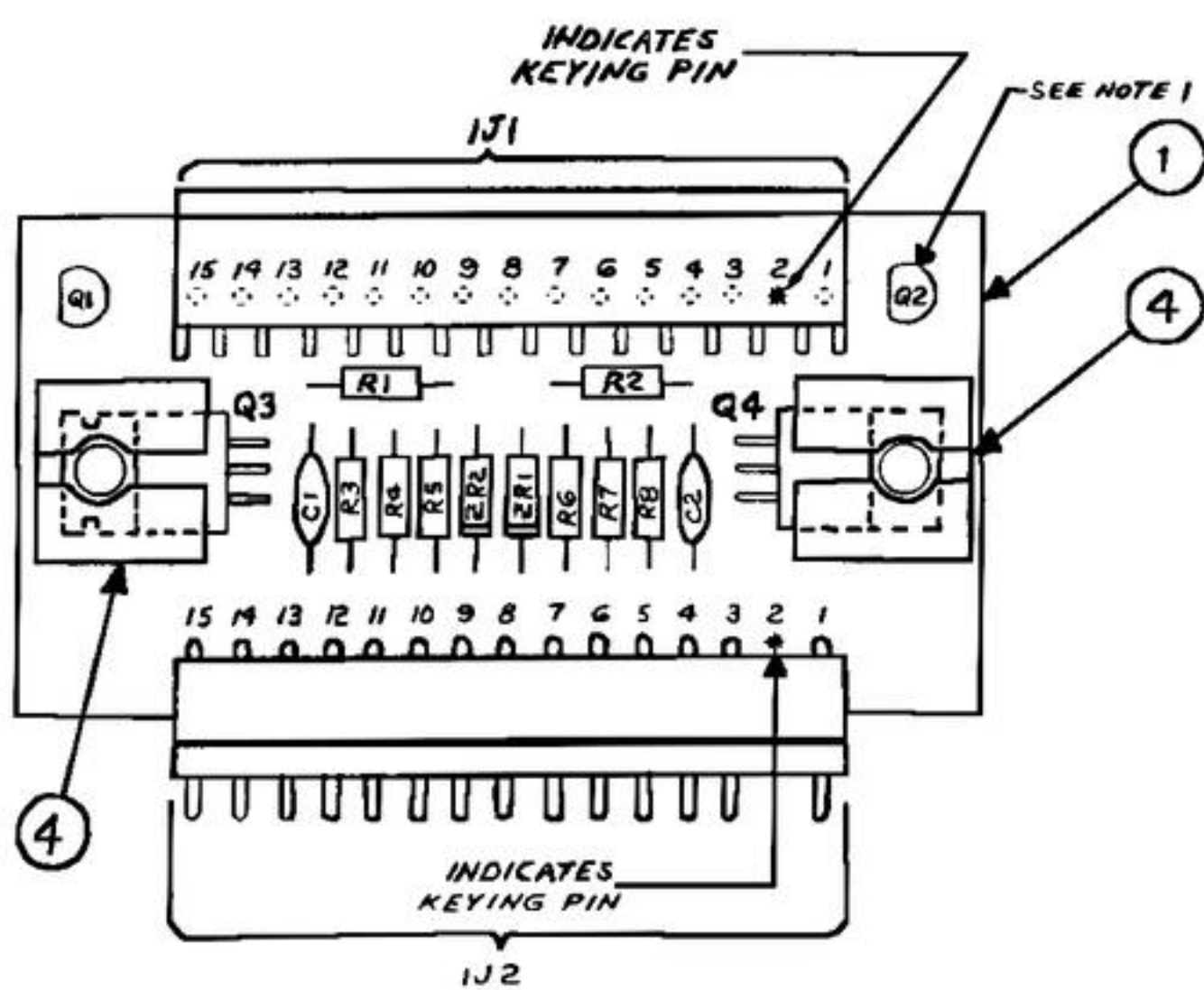
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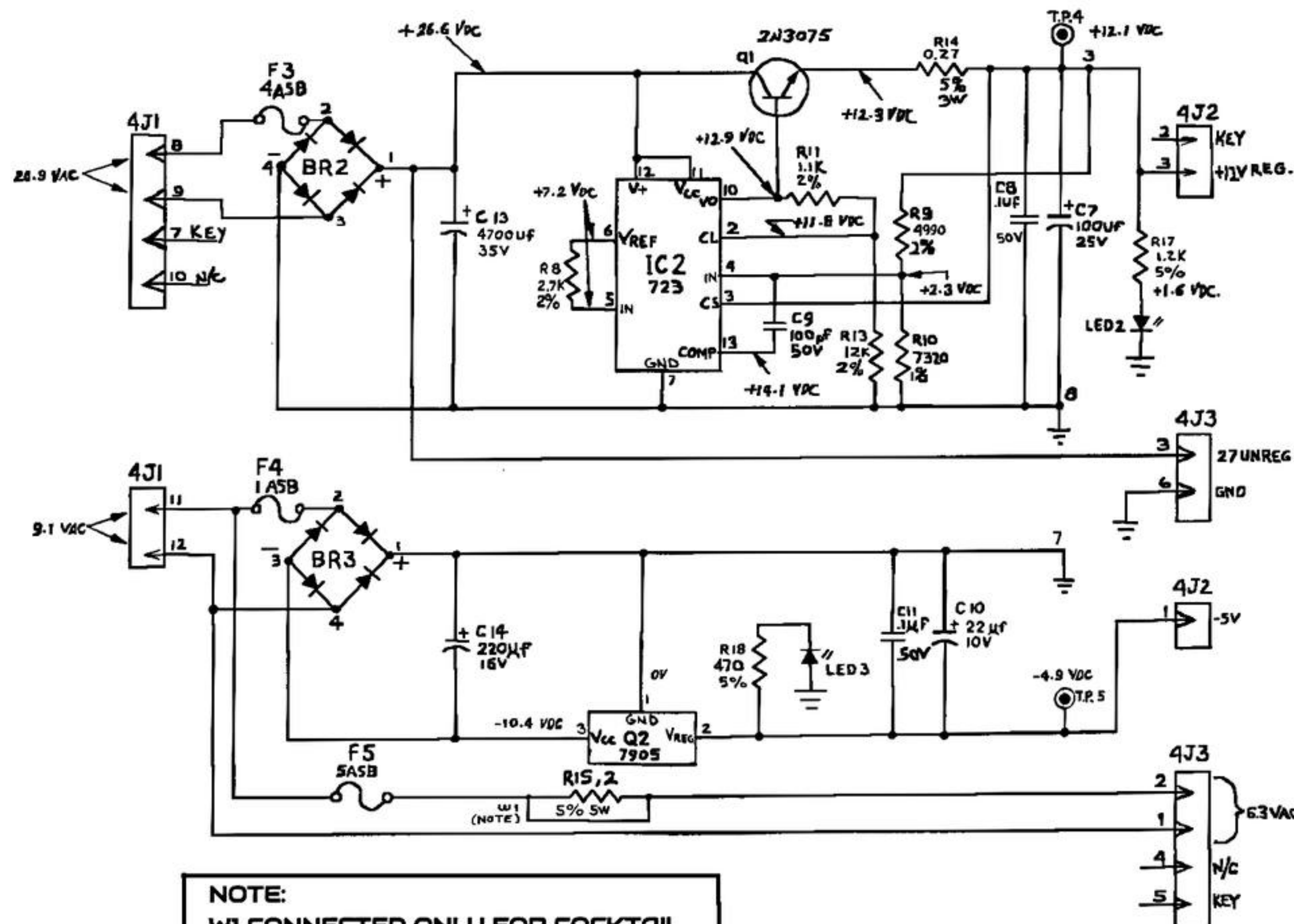
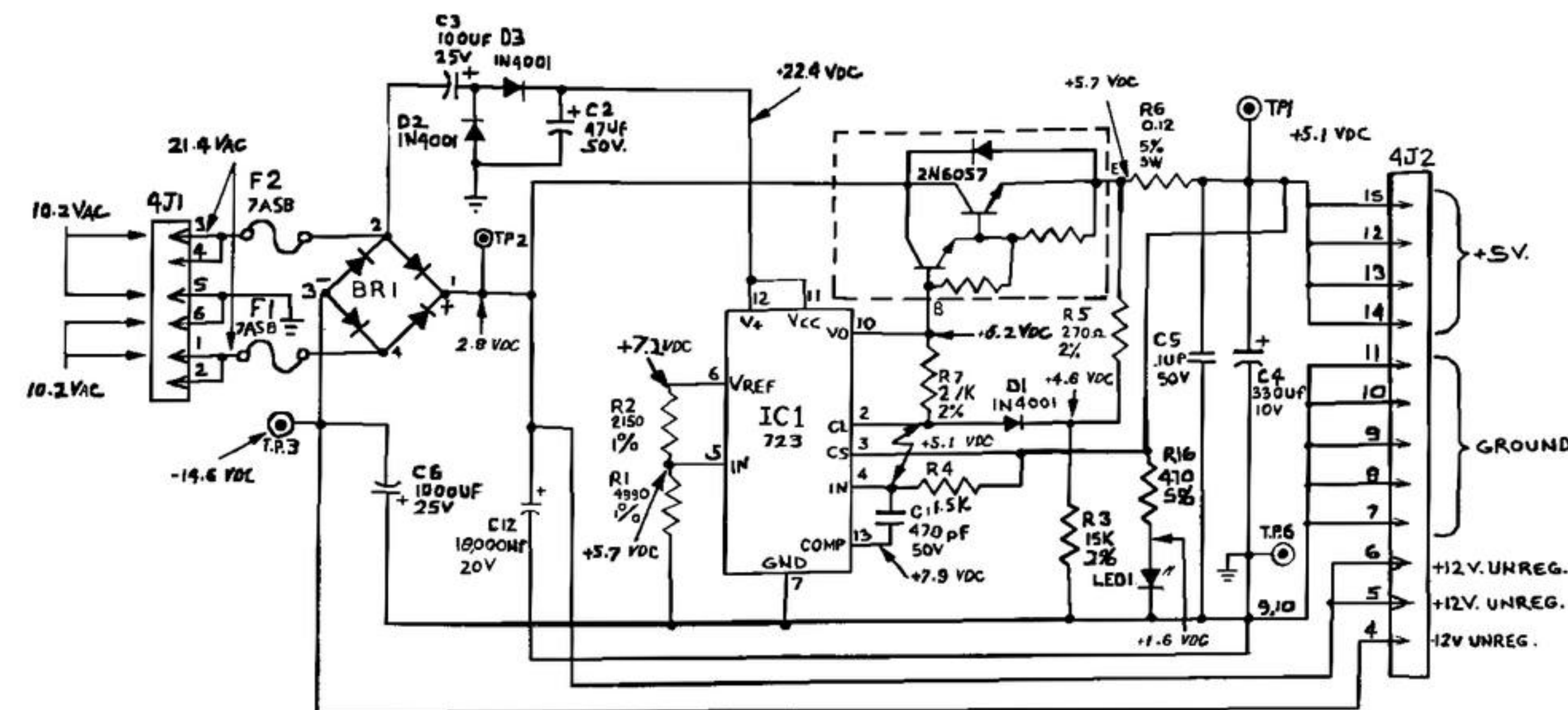
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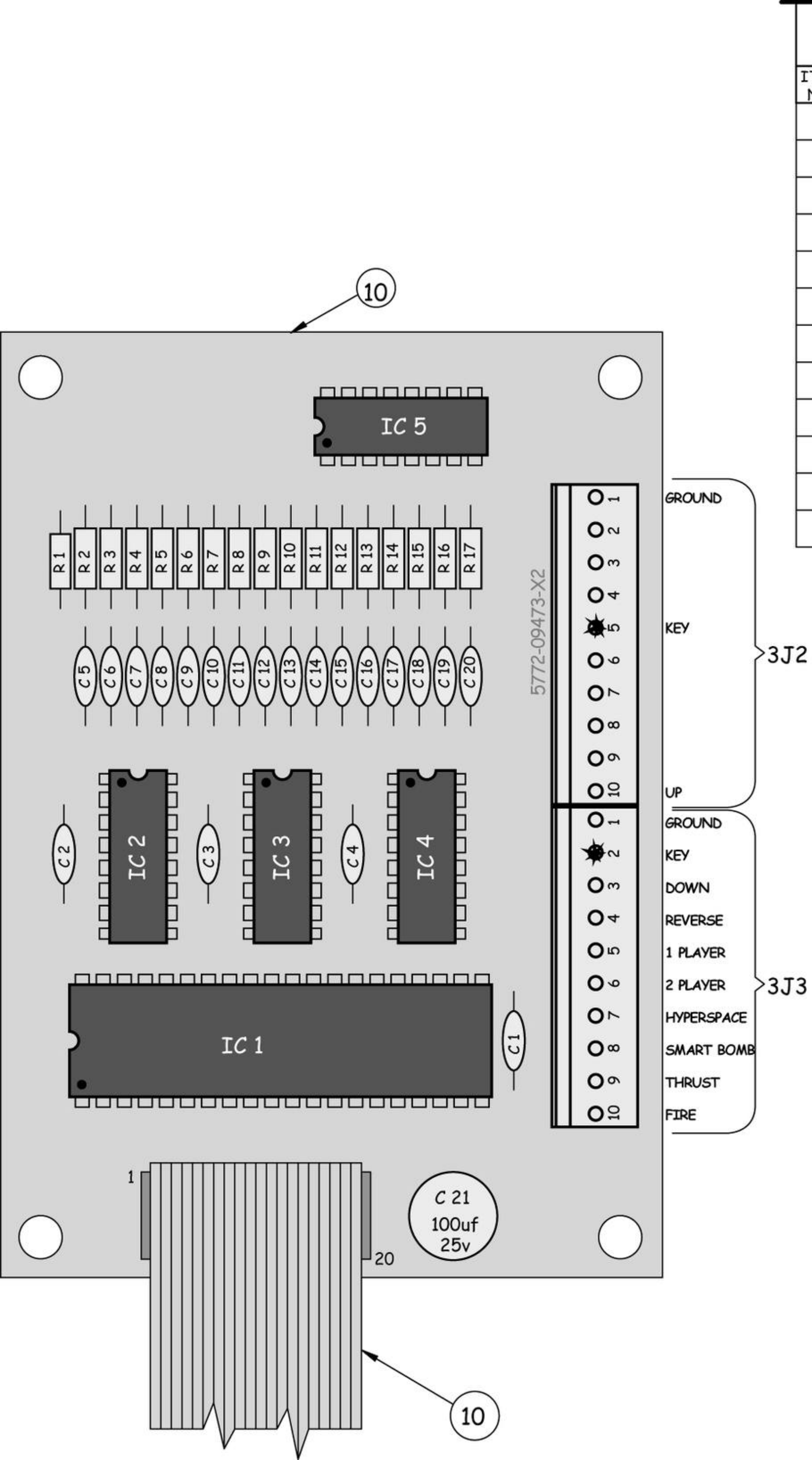
- NOTES:
1. OBSERVE POLARITY OF CAPACITORS DIODES AND POSITION OF TRANSISTORS.
 2. REFERENCE DRAWING: SCHEMATIC 16-8810

BILL OF MATERIALS				
ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D NO.
1	5773-09664-00		BARE P.C. BOARD	1
2	5792-09676-00	IJ1	RIGHT ANGLE CONNECTOR, 15 PIN	1
3	5791-09670-00	IJ2	HEADER CONNECTOR, 15 PIN	1
4	5705-09663-00		HEAT SINK, AAVID 5741	2
	5705-09688-00		HEAT SINK, THERMALLOY 6043	0
7	5130-09661-00	Q3, Q4	SCR 122F (100V)	2
8	5190-09430-00	Q1, Q2	TRANSISTOR MPS A55	2
9	5075-09662-00	ZR1	ZENER IN5244 ± 5% (14V)	1
10	5075-09406-00	ZR2	ZENER IN5234 ± 5% (6.2V)	1
11	5010-09036-00	R1, R2, R5, R6	RESISTOR C.F. 100 Ω ± 5% 1/4 W	4
12	5010-09039-00	R3, R8	RESISTOR C.F. 10 Ω ± 5% 1/4 W	2
13	5010-09434-00	R4, R7	RESISTOR C.F. 22 Ω ± 5% 1/4 W	2
14	5043-08996-00	C1, C2	CAPACITOR 0.1 μF CERAMIC +80% -20% 50V	2

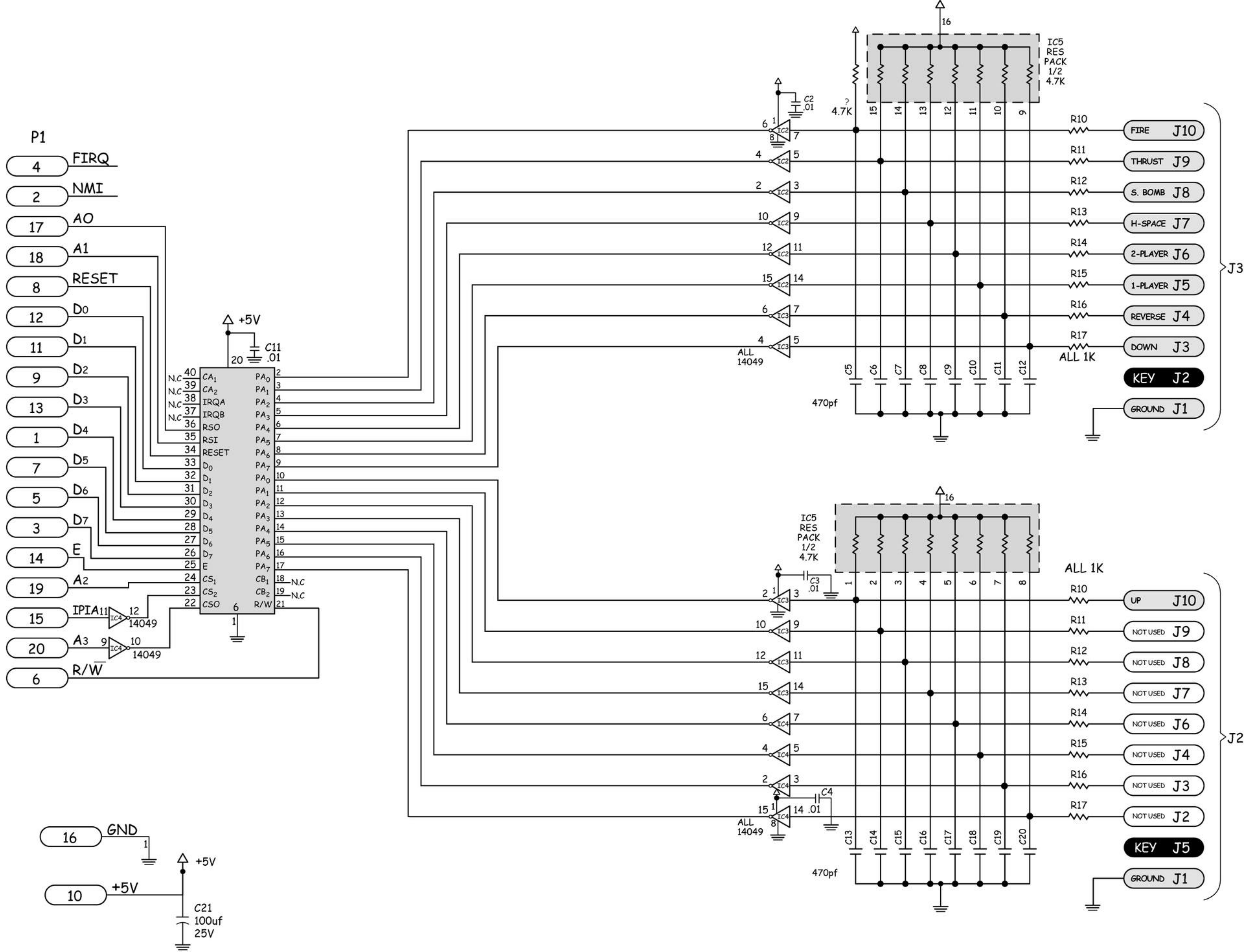


NOTE:
W1 CONNECTED ONLY FOR COCKTAIL APPLICATIONS.

Power Supply Board Schematic Diagram and
Overvoltage Protection Board



BILL OF MATERIAL						
ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D NO.	ALTERNATE PART NO.	ALTERNATE PART NO.
1	5772-09473-00	...	BARE P.C. BOARD	1	5772-09473-X2	
2	5310-08375-00	IC2, IC3, IC4	4049, INV. HEX BUFFER	3	...	...
3	5019-09239-00	IC5	4.7 K 15 RES, 16 DIP	1	4116R-2-472LF	916C472X2PE
4	5010-08331-00	R1	4.7K Ω RESISTER, 1/4W 5%	1	...	...
5	5043-05360-00	C1, C2, C3, C4	.01 MFD. CAPACITOR 50V. AXIAL, CERAMIC +80-20%	4	...	...
6	5043-05065-00	C5 THRU C20	470 PFD. CAPACITOR 50V. AXIAL, CERAMIC	16	...	...
7	5040-09421-00	C21	100 MFD. CAPACITOR 25V. RADIAL, ELECTROLYTIC	1	...	...
8	...	...		...		
9	5791-09444-00	J2 J3	10 PIN HEADER, MOLEX 09-65-1101	2	...	...
10	5795-09540-00	3P1	20 PIN RIBBON CABLE ASSY.	1	...	...
11	5430-08972-00	IC1	MC6821 P1A?	1	...	...
12	5010-09358-00	R2 THRU R17	1K OHM RESISTOR 1/4W 5%	16	...	...



J3

J2

GAME NAME		"DEFENDER" RED LABEL	DEFENDER BLUE LABEL	DEFENDER GREEN LABEL	DEFENDER WHITE LABEL	
ASSEM. NO.		D-8572-3001	D-8520-00	D-8520-00		
PART DESIGNATION	IC1	A-5343-09636	5343-09595-00	5343-09621-00		
	IC2	A-5343-09637	5343-09596-00	5343-09596-00		
	IC3	A-5343-09638	5343-09597-00	5343-09622-00		
	IC4	A-5343-09639	NOT USED	5343-09623-00		
	IC5	NOT USED	NOT USED	NOT USED		
	IC6	A-5343-09640	5343-09600-00	5343-09600-00		
	IC7	A-5343-09641	5343-09601-00	5343-09601-00		
	IC8	A-5343-09642	5343-09602-00	5343-09602-00		
	IC9	A-5343-09643	5343-09603-00	5343-09603-00		
	IC10	A-5343-09644	5343-09604-00	5343-09604-00		
	IC11	A-5343-09645	5343-09605-00	5343-09605-00		
	IC12	A-5343-09646	5343-09606-00	5343-09606-00		
JUMPERS		W2, W4, W6, W7, W9, W11, W13, W16, W17, W20, W21, W24	W2, W3, W6, W7, W10, W11, W13, W16, W17, W20, W21, W24	W1, W4, W6, W7, W10, W11, W13, W16, W17, W20, W21, W24		

PART NUMBER

Power SupplyParts List

PARTS LIST ARRANGED BY GROUPS OF RELATED PARTS			DRAWING DESIGNATION	PART NUMBER	DESCRIPTION
DRAWING DESIGNATION	PART NUMBER	DESCRIPTION	FUSES		
			F1, 2	5731-09432	Fuse, 7A, S.B.
			F3	5731-06314	Fuse, 4A, S.B.
			F4	5731-06569	Fuse, 1A, S.B.
			F5	5731-08665	Fuse, 2A, S.B.
			MISCELLANEOUS		
			TP2, 3, 5, 6	5824-09248	Terminal # 1502-1 (Test Point)
			LED 1, 2, 3	5671-09019	LED, Red
			--	5705-09543	Heat Sink
			--	20-09229-00	Thermal Compound, 0.02 Ounce
			--	4005-01051-07	Machine Screw, 5-40x7/16
			--	4405-01117	Hex Nut, 5-40
			--	5701-09538	Insulator, Mica, 3 mil
			--	03-7598	Insulator, Nylon
			--	03-7520	Tie Wrap
			--	5732-09178	Fuseholder
			A	B-8416	Heat Sink Assembly
			1A	5162-09425	Transistor, NPN, 2N6057
			2A	5700-09445	Transistor Socket, T03
			3A	5705-09431	Heat Sink
			4A	20-09229-00	Thermal Compound, 1.5 Ounce
			5A	5701-09538	Insulator, MICA, 3 Mil, T03
			6A	4006-01003-8	Machine Screw #6-32 x 1/2"
			7A	5853-09591	Lead Wire, #18 AWG, 4", 3/8" stripped (Green)
			8A	5853-09592	Lead Wire, #18 AWG, 4", 3/8" stripped (White)
			9A	5853-09593	Lead Wire, #18 Awg, 4", 3/8" stripped (Yellow)
INTEGRATED CIRCUITS					
IC1, 2	5460-09424	Voltage Regulator 723			
TRANSISTORS					
Q1	5161-09552	2N3055 Transistor, NPN			
Q2	5250-09515	Voltage Regulator 7905			
DIODES					
D1, 2, 3	5070-06258	Diode, 1N4001			
BR1	5100-09418	Bridge Rectifier, 35A, 100V			
BR2	5100-09513	Bridge Rectifier, 4.0A, 50V			
BR3	5100-09514	Bridge Rectifier, 1.0A, 50V			
RESISTORS					
R1, 9	5013-09427	Resistor, 4.99K Ohm, 1%, 1/4W			
R2	5013-09426	Resistor, 2.15K Ohm, 1%, 1/4W			
R3	5010-09428	Resistor, 1.5K Ohm, 2%, 1/4W			
R4	5010-09085	Resistor, 1.5K Ohm, 5%, 1/4W			
R5	5010-09508	Resistor, 270 Ohm, 2%, 1/4W			
R6	5012-09429	Resistor, 0.12 Ohm, 5%, 5W			
R7, R8	5010-09541	Resistor, 2.7K Ohm, 2%, 1/4W			
R10	5013-09542	Resistor, 7.32K Ohm, 1%, 1/4W			
R11	5010-09509	Resistor, 1.1K Ohm, 2%, 1/4W			
R12 not used					
R13	5010-09510	Resistor, 12K Ohm, 2%, 1/4W			
R14	5012-09512	Resistor, 0.27 Ohm, 5%, 5W			
R15	5012-09544	Resistor, 2.0 Ohm, 5%, 5W			
R16, 18	5010-09416	Resistor, 470 Ohm, 5%, 1/4W			
R17	5010-09314	Resistor, 1.2K Ohm, 5%, 1/4W			
CAPACITORS					
C1	5043-09065	Capacitor, 470 PFD, 50V, Axial			
C2	5040-09422	Capacitor, 47 MFD, 50V, Radial			
C3, 7	5040-09421	Capacitor, 100 MFD, 25V, Radial			
C4	5040-09423	Capacitor, 330 MFD, 10V, Radial			
C5, 8, 11	5043-08996	Capacitor, 0.1 MFD, 50V, Axial			
C6	5040-09420	Capacitor, 1000 MFD, 25V, Radial			
C9	5043-09492	Capacitor, 100 PFD, 50V, Axial			
C10	5040-09493	Capacitor, 22 MFD, 10V, Radial			
C12	5040-09419	Capacitor, 18,000 MFD, 20V, Radial			
C13	5040-09504	Capacitor, 4700 MFD, 35V, Radial			
C14	5040-09506	Capacitor, 220 MFD, 16V, Radial			

NC = NORMALLY
CLOSED

COM = COMMON
(GROUND)

NO = NORMALLY
OPEN

